DeCovarT: a Covariance-driven statistical framework to retrieve the cellular composition of bulk transcriptomic profiles

Bastien Chassagnol[1,2*†], Marielle Péré[3†], Mohamad Al Charif[1], Grégory Nuel[2], Etienne Becht[4], Elias Ventre[5], Anaïs Baudot[1]

[1]Marseille Medical Genetics (MMG), Systems Biomedicine team, Aix-Marseille University, Inserm U1251, Faculté de Médecine, 27 Bd Jean Moulin, Marseille, 13385, France.
[2]Laboratoire de Probabilités, Statistique et Modélisation (LPSM), UMR 8001, Sorbonne Université / CNRS, Paris, France.
[3]MacBes team, Inria Centre at Université Côte d'Azur; Institut de Pharmacologie Moléculaire et Cellulaire (IPMC), Sophia-Antipolis, France.
[4]Centre de Recherche sur l'Inflammation (CRI), UMR 1149, Inserm / Université Paris Cité, Faculté de médecine – site Bichat, 16 rue Henri Huchard, Paris, 75018, France.
[5]team COMPO (COMPutational pharmacology and clinical Oncology), Inria, Marseille, France.

*Corresponding author(s). E-mail(s): bastien_chassagnol@laposte.net;
Contributing authors: marielle.pere@inria.fr; nuel@math.cnrs.fr;
etienne.becht@inserm.fr; elias.ventre@inria.fr;
anais.baudot@univ-amu.fr;
†B.C. and M.P. contributed equally to this work.

Abstract

Introduction. Bulk transcriptomic analyses have contributed to the characterisation of tissues and diseases, but each profile averages the expression of every cell in the sample, so a shift in cell-type composition and a change of expression within a cell type are confounded. Single-cell assays separate those signals, yet their cost and the destruction of the sample limit them to a few specimens per condition. Computational deconvolution estimates the cell-type composition

of a bulk profile from reference profiles of purified populations or of annotated single cells. Current methods perform poorly on closely related cell populations, whose average profiles nearly coincide, and most return a point estimate without uncertainty. We hypothesised that the gene–gene covariance of each reference population, which encodes its gene-regulatory network, carries the information needed to separate them.

Methods. We developed DeCovarT, a generative model of the bulk profile in which the expression of each cell type is a latent multivariate Gaussian vector with its own mean and its own sparse precision matrix. The bulk is the sum of the cell-type profiles weighted by their relative abundances, so its law is a weighted Gaussian convolution with an explicit likelihood. Reference profiles supply the means; the cell-type covariances are estimated from single cells by the graphical lasso or by covariance shrinkage, and weights on the $\boldsymbol{\ell_1}$ penalty can carry prior mechanistic knowledge. We estimate the proportions by maximum likelihood with a Newton–Raphson algorithm, after a reparametrisation of the log-likelihood on the isometric log-ratio chart that enforces the simplex constraint exactly. The likelihood also supplies Wald and profile-likelihood intervals, and boundary likelihood-ratio tests for the absence of a cell type.

Results. In synthetic scenarios, we changed the overlap between cell-type profiles by changing only their gene–gene correlations. Larger overlap lowered the accuracy of standard deconvolution methods, including quadratic programming as implemented in *DeconRNASeq*, while DeCovarT had lower error in the scenarios we evaluated, most clearly when the cell-type profiles overlapped strongly.

Use case. Gastruloids are stem-cell aggregates that recapitulate early embryonic development, and their viability and morphology vary widely, even among structures grown with the same protocol. We apply DeCovarT to gastruloid bulk profiles with a single-cell reference, and use the estimated proportions to define a time-dependent potency score: how far the pluripotent cells present at a snapshot can still give rise to mature cell types.

Code and resources. Theory, graphical models and derivations are in the HAL preprint; the R package and its vignettes are on GitHub.

Keywords: deconvolution, bulk RNA-seq, weighted multivariate Gaussian distribution, covariance-driven model

1 Introduction

1.1 Bulk versus single-cell transcriptomics

- bulk RNA-seq returns one profile per sample, averaging measurements over several distinct cell populations
- that sum hides cell-specific signal, with two consequences
  - loss of specificity: a gene is called differentially expressed when only the fraction of the cell type producing it has increased
  - loss of sensitivity: genes of a minor type are masked by the variable expression of a major type

- gastruloids: morphotypes diverge along a trajectory of lineage fractions and cell states, even among structures grown with the same protocol [1]
  - viability and morphotype vary from batch to batch under matched conditions, which limits the wider use of organoids

    List the biological sources of batch-to-batch variability in gastruloid protocols (initial cell number, Wnt-agonist pulse timing, culture medium lot), each with a citation from [1] or its references.
  - early cell-type composition is a candidate predictor of the morphotype a gastruloid will reach
- single-cell RNA-seq resolves types, states and within-type spread at one snapshot
  - it names the driver of a failed structure: a missing lineage, a stalled state, or extra variance inside a type
  - cost and analysis burden limit cohorts to a few structures per condition
  - dissociation destroys the gastruloid and each structure yields few cells, so one gastruloid cannot be followed through time; every profile is a unique snapshot
- developmental cells lie on a continuum of differentiation with graded potency, unlike mature blood cells that fall into discrete lineages; neighbouring states then share most of their mean expression
- deconvolution bridges the two technologies: a single-cell reference supplies type-specific profiles, and cheap bulk samples supply the cohort

1.2 Related deconvolution methods

Every method below writes the bulk on a linear scale, $\boldsymbol{y}_{\cdot i} = \sum_j p_{ji} \boldsymbol{x}_{\cdot j,i}$. The methods differ in the law of the purified profiles, in whether genes are independent, and in whether the variance of type j enters the bulk with weight p_{ji} (independent cells) or p_{ji}^2 (one random population profile). Methods that infer cell-type-specific expression given the composition, and semi-reference and reference-free methods, are reviewed in Appendix C.

- regression on reference means is the baseline: *CIBERSORT* [2, 3], *DeconRNASeq* [4], *MuSiC* [5], *DWLS* [6], *Bisque* [7], *deMixSC* (Guo et al. 2025)
  - shared assumption: $\mathbb{E}[\boldsymbol{y}_{\cdot i}] = \boldsymbol{\mu p}_{\cdot i}$ with independent genes; profile variability enters, at most, as a fixed weight per gene
  - *MuSiC* weights gene g by the inverse of a between-subject variance; *DWLS* by the inverse squared fitted mean, capped by a dampening constant, so that rare and abundant types are estimated with comparable relative error; *Bisque* learns a gene-wise affine map from bulk to pseudo-bulk
  - *deMixSC* keeps weighted NNLS and combines the *DWLS* and *MuSiC* weights; a matched bulk and single-nucleus benchmark selects platform-sensitive genes, which are down-weighted rather than corrected
  - all these weighted least-squares fits are generalised-method-of-moments estimators: they solve sample moment conditions and return a point estimate, with no interval unless a variance model, a sandwich or a resampling unit is added

Add Guo et al. 2025 (*deMixSC*) to Zotero
and replace the link by a citation key.

  - no measurement model: least squares on intensities, often after TPM or CPM normalisation
  - that normalisation is a closure: the profile is divided by its own total, so the expression of the last gene is fixed by the others, the covariance becomes singular, and genes that were independent become negatively correlated [8]; for independent genes with variances v_g, normalised mean z_g and library size L, the delta method gives $\mathrm{Cov}(z_g, z_{g'}) \approx -(z_g v_{g'} + z_{g'} v_g - z_g z_{g'} \sum_h v_h)/L^2$ for $g \neq g'$ (Appendix D.3); the gene network estimated on raw counts is then no longer the network of the normalised bulk, and the target becomes transcript fractions
  - output: a point estimate on the simplex, from ν-support vector regression, quadratic programming or weighted non-negative least squares (NNLS) [9]
- univariate distribution-aware likelihoods: *RNA-Sieve* [10], *ReDeconv* [11], *RCTD* [12], *DeconV* [13]
  - the bulk is a sum of independent cells, so a type's variance enters linearly in p_{ji}; genes stay independent
  - the number of cells per observation decides the law: a spatial spot holds a handful of cells and a few hundred molecules, so the count distribution and its mean–variance relationship must be modelled explicitly for unbiased and consistent estimates; a bulk sample sums thousands to millions of cells, which puts each gene in the central-limit regime, and when the target is the mean composition with averaged cell-type moments plugged in, a Gaussian with the first two moments suffices; *RNA-Sieve*, *DECALS* and *ReDeconv* rely on this device [10, 11, 14]
  - *RNA-Sieve* and *ReDeconv* use a central-limit Gaussian per gene; *RCTD*, designed for spatial spots, a Poisson law with log-normal overdispersion and a gene-wise platform ratio $\tilde{b}_g = e^{\gamma_g}$ estimated in a first stage [12]; *DeconV* a Poisson bulk
  - *RNA-Sieve* models noisy reference means across several bulks and returns Fisher or Godambe confidence regions; *RCTD* and *ReDeconv* return point estimates
- multivariate, covariance-aware methods: *DECALS* [14] and *MEAD* [15]
  - *DECALS* assumes $\mathrm{Cov}(\boldsymbol{y}_{\cdot i}) = \sum_j p_{ji}^2 \boldsymbol{\Sigma}_j$, the first two moments of DeCovarT without a density; it estimates $\boldsymbol{\Sigma}_j$ by regressing residual outer products on $\hat{p}_{ji}^2$ across bulks, then returns constrained least squares with a sandwich covariance
  - *MEAD* is an errors-in-variables model with a sample scale and a random gene-wise platform ratio; it returns delta-method intervals as $G \to \infty$ under an almost-sparse dependence graph estimated from the bulk

  - both need many bulk samples of varying composition; in *MEAD* the gene dependence enters only the sandwich, not the point estimate
- Bayesian estimation of proportions: *DSection* [16], *BLADE* [17], *BEDwARS* [18], *BayesPrism* [19]
  - Dirichlet prior on $\boldsymbol{p}_{\cdot i}$, linear mixing, profiles drawn around the reference (Gaussian, log-normal or a single-cell prior)
  - posterior summaries by Gibbs, Metropolis–Hastings or variational inference; none models a $G \times G$ gene covariance

Most of these methods treat the reference signature as a fixed design. The bulk is regressed on $\boldsymbol{\mu}$ as in ordinary least squares, $\boldsymbol{y}_{\cdot i} = \boldsymbol{\mu}\boldsymbol{p}_{\cdot i} + \boldsymbol{\varepsilon}_i$, with an error assumed independent of the regressors and with a law that does not depend on $\boldsymbol{p}_{\cdot i}$: the exogeneity assumption of linear regression. In contrast to standard linear regression, DeCovarT relaxes exogeneity by treating latent cell-type expression profiles as random rather than fixed. The residual $\boldsymbol{y}_{\cdot i} - \boldsymbol{\mu}\boldsymbol{p}_{\cdot i} = \sum_j p_{ji}(\boldsymbol{x}_{\cdot j,i} - \boldsymbol{\mu}_{\cdot j})$ still has mean zero, but its covariance $\sum_j p_{ji}^2 \boldsymbol{\Sigma}_j$ depends on the unknown composition, so the noise itself carries information on $\boldsymbol{p}_{\cdot i}$. DeCovarT shares this view with *RNA-Sieve*, *DECALS*, *MEAD* and the Bayesian methods, but most fixed-signature engines use neither the within-reference (cell-to-cell) nor the between-reference (donor-to-donor) variability of the cell-type profiles.

Three gaps remain. First, no reference-based method uses a type-specific gene–gene covariance to shape the point estimate itself, which is where closely related cell types differ. Second, few methods return calibrated intervals for a single bulk sample, and none handles a proportion on the boundary of the unit simplex, where one or several cell types are absent (active null proportions). Third, estimating $\boldsymbol{p}$ jointly with $\{\boldsymbol{\Sigma}_j\}$ and the technical noise from the same bulk samples is weakly identified; unlike *MEAD*, *DECALS* and *RNA-Sieve*, which estimate some of these terms from the bulk cohort, DeCovarT plugs in $(\boldsymbol{\mu}, \{\boldsymbol{\Sigma}_j\})$ from the reference and estimates $\boldsymbol{p}$ only.

1.3 DeCovarT

- model: each cell-type profile is a multivariate Gaussian with a sparse precision (a gene-regulatory network), and the bulk is their weighted convolution, itself a multivariate Gaussian with an explicit likelihood [20]
- reference second moments: five estimators of the cell-type-specific covariance are compared on a single-cell reference (Section 3.7.3): partial correlations, the graphical lasso with an unpenalised diagonal, a de-sparsified graphical lasso with false discovery rate control, a Poisson–log-normal network, and a dense shrinkage covariance
- estimation: constrained maximum likelihood on the isometric log-ratio (ILR) chart of the simplex [21]
- inference: Wald and profile-likelihood intervals for each proportion, and boundary likelihood-ratio tests for the absence of one or several cell types (active null proportions) with chi-bar-square null laws [22]
- a synthetic simulator that controls the difficulty of each scenario

  - Gram construction of the mean signature to a target cosine
  - random network topologies for the precision of each type
  - a covariance scale tuned to a target MixSim overlap [23, 24]: the probability of misassigning a cell from its own profile, hence a lower bound on the error of any composition estimate built from those profiles
  - normalised Shannon entropy of $\boldsymbol{p}$ to control imbalance

1.4 Outline

- Section 2: two synthetic settings, then the gastruloid application
- Section 3: the generative model, inference, the simulator, and how a single-cell reference is turned into DeCovarT inputs
- Section 4: extensions, including a Bayesian formulation
- appendices: reviews of network inference and of deconvolution, maximum-likelihood theory, performance metrics and supplementary simulations

2 Results

2.1 Overview

- Figure 1: generative model, bulk constructions and the linear mixture
- Figure 2: synthetic means, precision graphs, overlap and scores
- claim to test: a shared mean is not enough to separate close types
- larger effect expected on interval coverage than on point bias
- separation is possible when means match and precision graphs do not
- rare types give wide intervals; pool them, or treat them as a residual component
- reporting frame: ADEMP (bias, RMSE, coverage, failure), no composite score [25]

2.2 Bivariate toy: two genes, two cell types

- purpose: check that gene–gene correlation changes the estimated fractions when only two genes and two cell types are observed
- one free ILR coordinate on the simplex; the likelihood is a curve, then mapped back to $\boldsymbol{p}$
- each purified profile is Gaussian; the bulk is their abundance-weighted sum, with no extra noise (2)
- $G = 2$, $J = 2$, $N = 500$ Monte Carlo bulks per scenario, 972 scenarios
- composition $\boldsymbol{p}$: balanced $(1/2, 1/2)$, moderately unbalanced $(17/20, 3/20)$, highly unbalanced $(99/100, 1/100)$
- centroid gap: small, $\boldsymbol{\mu}_{\cdot 1} = (20, 22)$ versus $\boldsymbol{\mu}_{\cdot 2} = (22, 20)$; large, $(20, 40)$ versus $(40, 20)$
- within-type correlation $\rho_j \in \{-0.8, -0.6, \ldots, 0.8\}$, set separately for each type

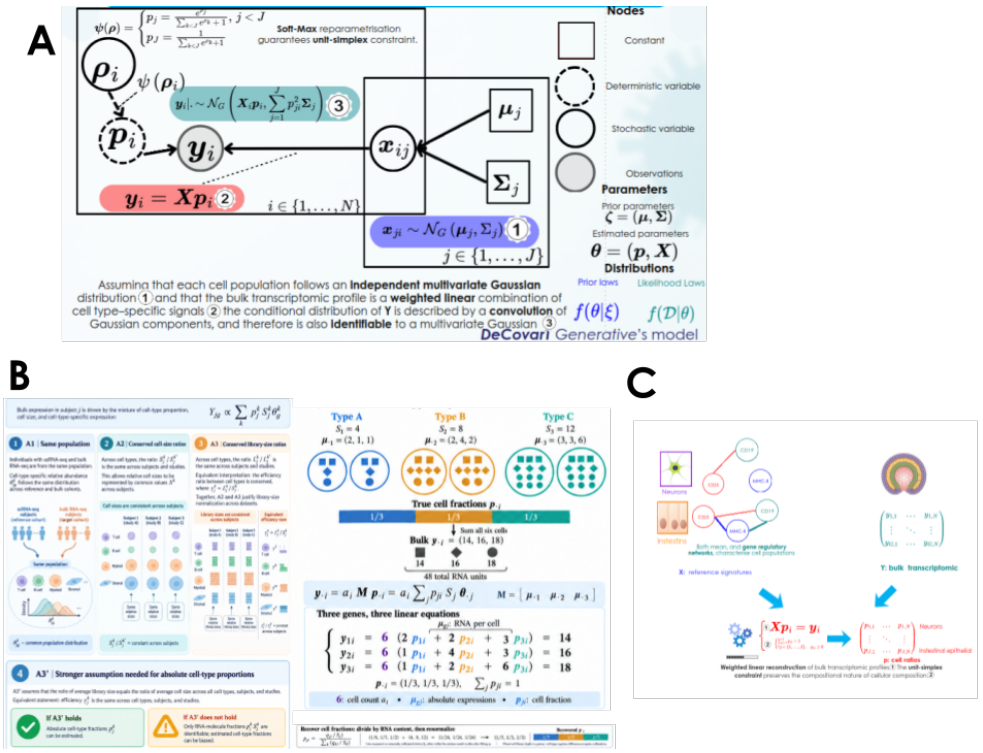


Fig. 1 Overview of DeCovarT. Top: plate diagram. The soft-max map ψ sends $\boldsymbol{\rho}_i$ to a composition $\boldsymbol{p}_i$. Latent profiles $\boldsymbol{x}_{\cdot j}$ are independent $\mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$. The bulk is the mixture $\boldsymbol{y}_i = \boldsymbol{X}\boldsymbol{p}_i$ and, marginally, the Gaussian convolution. Bottom left: three ways to build a bulk (one population; shared cell-size ratios; library-size ratios), and the further assumption required before $\boldsymbol{p}$ is a cell fraction. Bottom centre: three types, with RNA content μ_g times cell fraction p_j. Bottom right: reference means and gene-regulatory graphs, the bulk matrix, and the same linear reconstruction.

- marginal variances: homoscedastic ($\sigma^2 = 1$ for both genes) or heteroscedastic ($\sigma_1^2 = 1$, $\sigma_2^2 = 2$).
- solvers: NNLS; *DeconRNASeq* (`lsei`); L-BFGS-B; projected gradient; Newton–Raphson; Marquardt–Levenberg; simulated annealing

  keep only newton raphson, relocate others in appendix but add mead, decals and rna-sieve

- mean-only solvers minimise $\|\boldsymbol{y} - \boldsymbol{\mu}\boldsymbol{p}\|^2$ and carry no convolution Wald standard error
- code and outputs: script, vignette, plot books; file-by-file reading in Appendix G.1

From this simplest synthetic setting, we observe that the overlap was a good proxy of the quality of the estimation: the less the overlap between the two cell distributions, the better the quality of the estimation.

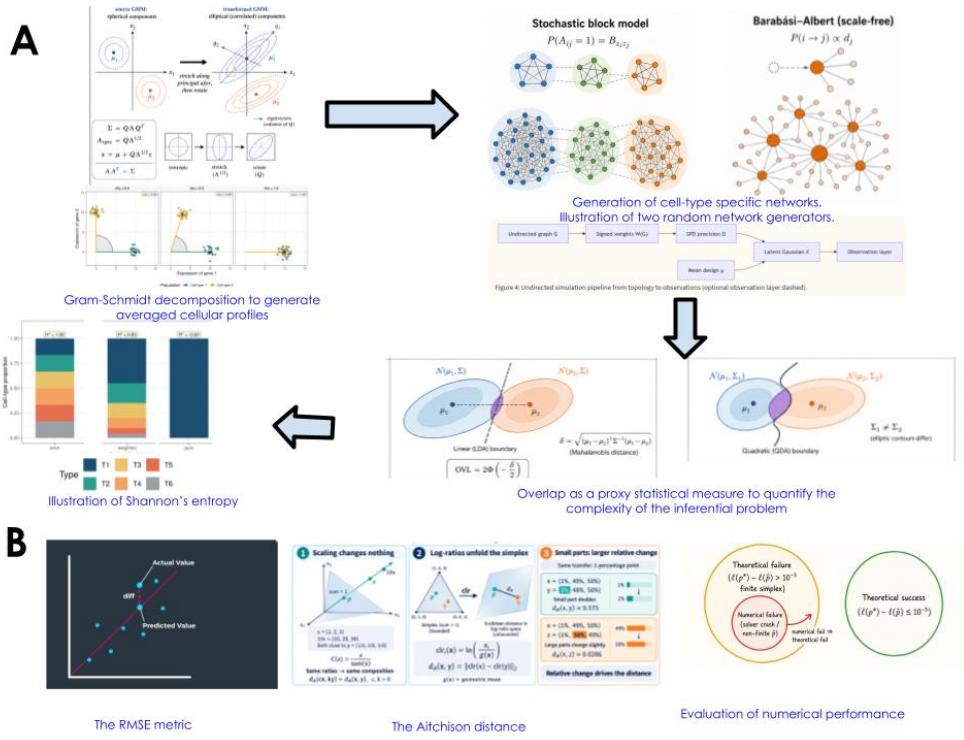


Fig. 2 Synthetic simulation framework. Panel A, left to right and top to bottom: Gram–Schmidt / spectral construction of the mean profiles; stochastic block model and Barabási–Albert precision graphs; Shannon entropy of $\boldsymbol{p}$ across six types; overlap of two purified Gaussians (linear and quadratic decision boundaries). Panel B: root-mean-square error, Aitchison distance on the simplex, and the split between theoretical failure of the likelihood and numerical failure of the solver.

2.3 Covariance-driven scenario: twenty genes, three cell types

The bivariate toy isolates the effect of gene–gene covariance in a fully transparent setting but cannot demonstrate scalability or the realistic topology of gene-regulatory networks (GRNs). We therefore designed a second scenario with $G = 50$ genes and $J = 3$ cell types (Figure 4). Specifically, since cell types 1 and 2 are near mean-collinear (cosine value of $\rho_{12} \approx 0.97$), mean-only methods cannot separate them; the network topology encoded in $\boldsymbol{\Sigma}_j$ is the only discriminative signal available to DeCovarT.

- role: two types with nearly collinear means, so the discriminative signal sits in the precision graphs (design in Section 3.6)
- $G = 20$, $J = 3$, seed `20260807`
- one Gram: cosines 0.9 (types 1–2) and 0.1 (either of those versus type 3); scale $s_\mu = 10$, so the Euclidean gaps are about 4.5 and 13.4
- types 1 and 2: scale-free or stochastic block model graphs, 2×2 assignments [26, 27]; type 3: scale-free, held fixed

- Strict positive definiteness is ensured by shifting the precision matrices applying the affine spectrum transformation (B1) [28].
- We impleemented a custom method to choose the covariance scale such that the averaged overlap hits 0.02, 0.08 or 0.20, matched at the balanced composition; unbalanced $\boldsymbol{p}$ raises the realised overlap while constraining the topological sutrcture (the zeros of the precision matrix)
- compositions from normalised Shannon entropy $H^\star \in \{1, 0.5, 0.1\}$: $(1/3, 1/3, 1/3)$, $(0.84, 0.08, 0.08)$ and $(0.98, 0.01, 0.01)$; $4 \times 3 \times 3 = 36$ scenarios
- solvers: *DeconRNASeq* (`lsei`); *CIBERSORT*; L-BFGS-B; Newton–Raphson; Marquardt–Levenberg

Replicates: the vignette states $N = 50$ bulks per scenario, `scripts/fig03_variance_driven.R` uses $N = 100$. Align both before reporting.

Planned benchmark: add *RNA-Sieve* [10], *DECALS* [14], *MEAD* [15] and *ReDeconv* [11] to the bivariate and covariance-driven grids, with the same descriptors and the metrics of Appendix F. *DECALS* and *MEAD* need several bulks per fit: give them the N replicates of one scenario as a cohort, and report that their design differs from a one-sample fit.

2.4 Biological application: gastruloid development

- dataset: murine gastruloids with a multimodal characterisation at 48, 72 and 96 hours [1]; single-cell reference and bulk profiles described in Section 3.7.1
- why gastruloids test deconvolution
  - high batch-to-batch variability of viability and morphotype under matched protocols
  - developmental states form a potency gradient, so adjacent states share most of their mean expression; mean-only engines are expected to confuse them
  - one dominant population at early stages, with several rare types
- comparison set: DeCovarT against the mean-only and distribution-aware engines of Section 1.2, on the same genes and the same reference
- ablation design, run as a factorial on pseudo-bulks of known composition built from held-out cells
  1. gene-selection stringency: the presence filter plus the cell-type main-effect test of Section 3.7.2, at increasing effect-size thresholds that yield nested panels from a few dozen to several thousand genes
  2. covariance or precision estimator: the five strategies of Section 3.7.3 (partial correlations, graphical lasso with an unpenalised diagonal, de-sparsified graphical lasso, Poisson–log-normal network, dense shrinkage), plus the diagonal as a mean-only control
  3. single-cell to bulk alignment: the model of Section 3.7.4 against the released quadratic-weight model and the alternatives of Appendix D.3

4. engines: DeCovarT and the comparison set, each at every panel size

- expected pattern, to be tested
  - too few genes: every engine fails, because the mean signature is rank deficient for close types and the covariance has too few coordinates
  - many genes, with noisy and platform-dependent genes removed: engines converge to similar accuracy, because the mean signature alone becomes injective
  - an intermediate range, or scenarios with strongly correlated types, where covariance-driven engines outperform mean-only ones
  - the result is a calibration curve of accuracy against panel size, stratified by the overlap of the purified types

Run the ablation; report each stage against the overlap and entropy descriptors of Section 3.6, not only the panel size.

Potency score: define it from the estimated proportions and validate it against the morphotype annotations.

3 Methods

3.1 Notation

Bulk sample i is fixed inside a derivation. The index i is restored in the model statements. Base indices used throughout: gene $g = 1, \dots, G$, cell type $j, \ell = 1, \dots, J$, sample $i = 1, \dots, N$, reference donor r, cell c. Table 1 is the alignment layer used in the single-cell alignment vignette.

3.2 Generative framework and core modelling assumptions

Set-up.

- bulk expression $\boldsymbol{y} = (y_{gi}) \in \mathbb{R}_+^{G \times N}$, with profile $\boldsymbol{y}_{\cdot i} \in \mathbb{R}_+^G$ for sample i; samples are independent
- mean signature $\boldsymbol{\mu} = (\mu_{gj}) \in \mathbb{R}^{G \times J}$; as in *MuSiC* [5], $\boldsymbol{\mu}_{\cdot j}$ stores the mean profile of cell type j across reference samples or cells, not a single draw
- unknown proportions $\boldsymbol{p} = (p_{ji})$, with composition $\boldsymbol{p}_{\cdot i}$ of sample i
- covariance $\boldsymbol{\Sigma}_j \in \mathbb{R}^{G \times G}$ and precision matrix $\boldsymbol{\Omega}_j \equiv \boldsymbol{\Sigma}_j^{-1}$ of cell type j: one network per type, stacked into a $G \times G \times J$ tensor
- most deconvolution models write the bulk as the linear combination of the mean profiles, with each composition on the unit simplex, the setting of constrained least-squares simplicial regression [29]:

$$\boldsymbol{y}_{\cdot i} = \boldsymbol{\mu}\, \boldsymbol{p}_{\cdot i}, \qquad \sum_{j=1}^{J} p_{ji} = 1, \qquad p_{ji} \geq 0 \quad (i = 1, \dots, N) \tag{1}$$

Table 1 Cell counts, single-cell moments, scales, and the two covariance mixtures. A tilde marks a quantity rescaled to a reference unit: per cell ($\tilde{\boldsymbol{y}}$, $\tilde{\boldsymbol{\mu}}$) or relative to the reference protocol ($\tilde{\boldsymbol{b}}$). $\boldsymbol{\rho}$ elsewhere in the paper is the isometric log-ratio coordinate of $\boldsymbol{p}$, so the within-type correlation is ϱ.

Symbol	Meaning
$m_{ji} \in \mathbb{N}$	Number of cells of type j in bulk sample i (*MuSiC* writes m_{kj}).
$n_i = \sum_j m_{ji}$	Total number of cells in bulk sample i.
$\boldsymbol{p}_{\cdot i} = \boldsymbol{m}_{\cdot i}/n_i$	Cell fractions, with $\boldsymbol{m}_{\cdot i} = (m_{1i}, \dots, m_{Ji})^\top$.
$\tilde{\boldsymbol{y}}_{\cdot i} = \boldsymbol{y}_{\cdot i}/n_i$	Bulk expression per cell.
$\tilde{\boldsymbol{\mu}}_{\cdot j} = C_j^{-1} \sum_{c \in j} \boldsymbol{x}_{\cdot c}$	Per-cell mean of the raw UMI counts $\boldsymbol{x}_{\cdot c}$ of the C_j reference cells of type j, without library-size normalisation.
$S_j = \mathbf{1}^\top \tilde{\boldsymbol{\mu}}_{\cdot j}$, $\boldsymbol{\theta}_{\cdot j} = \tilde{\boldsymbol{\mu}}_{\cdot j}/S_j$	Mean transcriptome size of a type-j cell and its relative abundance profile.
$q_{ji} = p_{ji} S_j / \sum_\ell p_{\ell i} S_\ell$	Transcript fractions.
a_i	Sample-wide scale between the reference and the bulk library.
b_g, $\tilde{b}_g = b_g^{\text{bulk}}/b_g^{\text{ref}}$	Capture efficiency of gene g in one protocol, and the gene-wise platform ratio, $\tilde{\boldsymbol{b}} = (\tilde{b}_1, \dots, \tilde{b}_G)^\top$.
$\boldsymbol{D}_i = \boldsymbol{\Sigma}_{\text{obs},i} = \text{diag}(\tau_{1i}^2, \dots, \tau_{Gi}^2)$	Diagonal, heteroscedastic bulk measurement noise.
$\boldsymbol{T}_{ji} = \sum_{c=1}^{m_{ji}} \boldsymbol{X}_{jic}$	Total expression contributed by type j to sample i.
$\boldsymbol{\Sigma}_{W,j}$, $\boldsymbol{\Sigma}_{B,j}$	Within-donor cell-to-cell covariance and between-donor covariance of type j.
$\boldsymbol{S}_W(\boldsymbol{v}) = \sum_j v_j \boldsymbol{\Sigma}_{W,j}$	Linear covariance mixture with weights $\boldsymbol{v}$.
$\boldsymbol{S}_B(\boldsymbol{v}) = \sum_j v_j^2 \boldsymbol{\Sigma}_{B,j}$	Quadratic covariance mixture.
$\varrho_j \in [-1/(m_{ji}-1), 1]$	Cell-to-cell correlation inside type j.
$\boldsymbol{\vartheta}$	Generic parameter vector.
$\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \boldsymbol{\mu}_{kl}, \boldsymbol{\Sigma}_{kl}$	First and second partial derivatives of the mean and covariance in ϑ_k, ϑ_l.

Core modelling assumptions.

(A1) linear mixing on a linear scale: $\boldsymbol{y}_{\cdot i} = \sum_j p_{ji}\, \boldsymbol{x}_{\cdot j,i}$, with counts or intensities, never logarithms

(A2) random cell-type profiles: $\boldsymbol{x}_{\cdot j,i} \sim \mathcal{N}_G(\boldsymbol{\mu}_{\cdot j}, \boldsymbol{\Sigma}_j)$, which relaxes exogeneity (Section 1.2)

(A3) independence of the profiles across cell types, so the bulk law is a Gaussian convolution

(A4) plug-in reference moments $(\boldsymbol{\mu}, \{\boldsymbol{\Sigma}_j\})$, each $\boldsymbol{\Sigma}_j$ positive definite

(A5) no platform scale and no extra measurement noise in this ideal model; both are added in Section 3.7.4

(A6) one fit per bulk sample

Under (A1)–(A4), the implied law is the estimating equation of DeCovarT; derivatives, the ILR chain rule and the Fisher information are in Appendix D.1 and the generative-model vignette.

$$\boldsymbol{y}_{\cdot i} \sim \mathcal{N}\left(\boldsymbol{\mu}\boldsymbol{p}_{\cdot i},\ \sum_{j=1}^{J} p_{ji}^2\, \boldsymbol{\Sigma}_j\right) \tag{2}$$

with log-likelihood, for one sample and $\boldsymbol{\Sigma}(\boldsymbol{p}) = \sum_j p_j^2 \boldsymbol{\Sigma}_j$,

$$\ell(\boldsymbol{p}) = C - \frac{1}{2}\log\det\boldsymbol{\Sigma}(\boldsymbol{p}) - \frac{1}{2}(\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Sigma}(\boldsymbol{p})^{-1}(\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p}) \tag{3}$$

- the factor p_{ji}^2 is the variance of a scaled random population profile; $\boldsymbol{S}_B(\boldsymbol{p}_{\cdot i})$ in Table 1 is this quadratic mixture
- plug-in $(\boldsymbol{\mu}, \{\boldsymbol{\Sigma}_j\})$; the unknown is $\boldsymbol{p}_{\cdot i}$ only
- the determinant term penalises compositions that inflate the covariance, while the Mahalanobis term is a generalised least-squares residual whose weight matrix moves with $\boldsymbol{p}$
- a sparse $\boldsymbol{\Omega}_j = \boldsymbol{\Sigma}_j^{-1}$ encodes a Gaussian Markov graph [28, 30], but the bulk precision $(\sum_j p_{ji}^2 \boldsymbol{\Sigma}_j)^{-1}$ is dense in general
- identifiability: $\boldsymbol{p}$ is identified when $\boldsymbol{p} \mapsto (\boldsymbol{\mu}\boldsymbol{p}, \sum_j p_j^2 \boldsymbol{\Sigma}_j)$ is injective; $\operatorname{rank}(\boldsymbol{\mu}) = J$ is sufficient but not necessary, since types with collinear means stay identified when their covariances differ
- Table 2 compares these assumptions with the closest likelihood- and moment-based methods

3.3 Inference

- objective: the Gaussian log-likelihood (3); with $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{\mu}\boldsymbol{p}$ and $\boldsymbol{\Omega} = \boldsymbol{\Sigma}(\boldsymbol{p})^{-1}$, its score is

$$\frac{\partial \ell}{\partial p_j} = -p_j \operatorname{tr}(\boldsymbol{\Omega}\boldsymbol{\Sigma}_j) + \boldsymbol{\mu}_{\cdot j}^\top \boldsymbol{\Omega}\boldsymbol{r} + p_j\, \boldsymbol{r}^\top \boldsymbol{\Omega}\boldsymbol{\Sigma}_j \boldsymbol{\Omega}\boldsymbol{r} \tag{4}$$

- the three terms come from the log-determinant, the generalised least-squares residual and the composition-dependent weight; the root of the score has no closed form, so the maximum is approximated by iterative second-order methods
- the analytic Hessian is in Appendix D.1; both derivatives were checked against numerical derivatives with Richardson extrapolation [31]
- keeping the factor $\frac{1}{2}$ of the Gaussian log-density on both terms of (3) makes the expected negative Hessian equal to the Fisher information used for the Wald intervals
- the score and Hessian above are ambient likelihood and Hessian: they ignore the simplex constraint, and the solvers work on a reparametrised log-likelihood instead (Section 3.4, Appendix D.2)
- default chart: isometric log-ratio (ILR) chart, $\boldsymbol{p} = \operatorname{softmax}(\mathbf{V}\boldsymbol{\rho})$ with a Helmert basis $\mathbf{V}$, a C^2 diffeomorphism of the open simplex [21]; additive log-ratio kept as an alternative chart

Table 2 Assumptions of DeCovarT and of the closest distribution-aware methods. The variance weight is p_j when the bulk sums independent cells and p_j^2 when each type contributes one random population profile.

Method	Profile law and gene dependence	Weight	Reference moments	Samples and output
RNA-Sieve [10]	central-limit Gaussian per gene; independent genes	p_j	noisy means, estimated jointly	several bulks; Fisher or Godambe regions
ReDeconv [11]	Gaussian per gene with transcriptome-size normalisation; independent genes	p_j	plug-in	one bulk; point estimate
DECALS [14]	first two moments, full $G \times G$ covariance per type	p_j^2	covariances estimated from the bulks	many bulks of varying composition; sandwich intervals
MEAD [15]	errors-in-variables Gaussian; almost-sparse global dependence	global	noisy means; random gene-wise platform ratio	many bulks; delta-method intervals as $G \to \infty$
DeCovarT	Gaussian with a sparse precision per type	p_j^2 (p_j in (5))	plug-in from the reference	one bulk; Wald, profile and boundary tests

- solvers
  - Newton–Raphson (default): analytic score and Hessian in ILR coordinates, with the trust-region safeguards of `stats::nlminb`
  - Marquardt–Levenberg (alternative): damped Newton (`marqLevAlg`), stopped on the relative distance to the maximum [32–34]
- the log-likelihood is not concave, also in the ILR chart; a converged solver certifies a local maximum at best, hence Dirichlet multistarts (Appendix H.1)
- one bulk column is an $N = 1$ fit: the interval describes that maximiser, and the maximum-likelihood bias is of order N^{-1}

3.4 Why the isometric log-ratio chart

- four charts map free coordinates onto the open simplex: the softmax of J scores, the additive log-ratio (ALR) to a reference type, the centred log-ratio (CLR) and the ILR [8, 35]; derivations and a side-by-side comparison are in Appendix D.2 and the generative-model vignette
- same statistics: the likelihood depends on $\boldsymbol{p}$ only, so the maximiser, the maximised log-likelihood and the delta-method covariance of $\hat{\boldsymbol{p}}$ are the same in every chart

- different numerics: the charts differ in the geometry seen by the solver, that is, gradient norms, step safeguards and the condition number of the Hessian, which sets the accuracy of every linear solve
- why ILR
  - it is the only one of the four charts that is minimal ($J-1$ free coordinates), symmetric in the cell types and an isometry for the Aitchison distance: a log-ratio between two types has the same length whatever their abundance, so a rare type weighs as much as an abundant one [21, 35]
  - its stationary points are exactly the interior Karush–Kuhn–Tucker (KKT) conditions points of the Lagrangian of the simplex problem: the chart absorbs the multiplier of $\sum_j p_j = 1$, and in the interior the inequality multipliers are zero by complementary slackness, so no constraint is left to handle [36]
  - it is numerically invariant: another orthonormal basis is an orthogonal rotation $\boldsymbol{\rho}^\star = \boldsymbol{Q}^\top \boldsymbol{\rho}$, which keeps the Hessian eigenvalues; a change of ALR reference is an oblique map whose condition number is $\sqrt{J}$, and it divides the curvature along the reference's own direction by up to J, which hurts when the reference type is rare or poorly identified; the free softmax adds a redundant direction along $\mathbf{1}$ with a zero Hessian eigenvalue, and CLR keeps J coordinates for $J-1$ degrees of freedom
- cost: a zero proportion lies at infinity in ILR coordinates, so faces of the simplex need the restricted fits of Appendix D.5

3.5 Confidence intervals for interior compositions

- scope: every $\hat{p}_j$ strictly positive; estimates on a face or a vertex of the simplex, and boundary tests for absent types, are treated in Appendix D.5
- Wald: expected Fisher information $I(\boldsymbol{p})$ (Appendix D.1) pulled back to ILR coordinates, $I_{\boldsymbol{\rho}} = \mathbf{J}_\psi^\top I(\boldsymbol{p}) \mathbf{J}_\psi$ with $\mathbf{J}_\psi = (\operatorname{diag} \boldsymbol{p} - \boldsymbol{p}\boldsymbol{p}^\top)\mathbf{V}$, then the delta method back to $\boldsymbol{p}$ [37]; symmetric, and it can leave $[0, 1]$ for a rare type
- intervals computed in ILR or logit coordinates and mapped back respect the bounds
- profile likelihood: χ^2_1 on the deviance [38], invariant to the chart and inside the simplex; preferred near a face, where the Wald statistic can shrink as the estimate moves away from the null (Hauck–Donner effect) [39]

> Planned: intervals from the generalised-linear-model view of DeCoVarT. Compute the $1 - \alpha/2$ and $\alpha/2$ quantiles in ILR or logit coordinates, where the quasi-score $\boldsymbol{D}^\top \boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{m}) = \mathbf{0}$ has information $\boldsymbol{D}^\top \boldsymbol{V}^{-1} \boldsymbol{D}$ (Section 4.2.4), then map both bounds back to the simplex. Compare their coverage with the delta-method intervals, which currently exceed the unit simplex for highly unbalanced mixtures (Appendix D.5.2).

- full argument: maximum-likelihood vignette

3.6 Synthetic scenarios

- purified profiles $\boldsymbol{x}_{\cdot j} \sim \mathcal{N}_G(\boldsymbol{\mu}_{\cdot j}, \boldsymbol{\Sigma}_j)$ are combined by the convolution (2), not drawn from a finite mixture; five layers are generated separately (pipeline in the synthetic-scenarios vignette)
- mean signature with a target Gram matrix (Figure 2A)
  - target cosines in R, by default equicorrelation $R_J(\rho) = (1-\rho)\mathbf{I}_J + \rho\mathbf{1}\mathbf{1}^\top$
  - $\boldsymbol{\mu} = s_\mu \mathbf{Q} R^{1/2}$ with an orthonormal frame $\mathbf{Q}$ and the symmetric spectral root, so $\boldsymbol{\mu}^\top\boldsymbol{\mu} = s_\mu^2 R$ exactly; a disjoint-support frame gives $\boldsymbol{\mu} \geq 0$ for quadratic-programming baselines
- precision graphs
  - skeletons: preferential-attachment scale-free [26], stochastic block model [27], small-world [40], Erdős–Rényi or hub graphs; scale-free graphs stress degree heterogeneity and are not a biological default [41]
  - signed weights $\pm v$ with an inhibitory fraction π_-, then a spectral shift $\boldsymbol{\Omega} = \boldsymbol{W} + (|\lambda_{\min}(\boldsymbol{W})| + u)\mathbf{I}$ that keeps the support and signs
- global covariance scale s: replacing $\boldsymbol{\Sigma}_j$ by $s\boldsymbol{\Sigma}_j$ keeps every structural zero and partial correlation, and only shrinks the Mahalanobis separation of the means; the covariance share of the Fisher information rises with s
- composition: one-dominant family $(1-(J-1)q, q, \ldots, q)$ solved for a target normalised entropy $H^\star(\boldsymbol{p}) = -\sum_j p_j \log p_j / \log J$
- scenario at a target overlap $\bar{\omega}^\star$
  1. fix $\boldsymbol{\mu}$, the graphs, signs, weights and cushion, giving base covariances $\boldsymbol{\Sigma}_j^{(0)}$; calibrate at the barycentre
  2. solve $\bar{\omega}(s) = \bar{\omega}^\star$ on $s \in [10^{-3}, 10^3]$ by Brent's method, with the same Monte Carlo points at every evaluation (common random numbers) so the objective is deterministic in s
  3. return $s^\star\boldsymbol{\Sigma}_j^{(0)}$; held fixed are the means, supports, signs and partial correlations
- overlap estimator
  - pairwise MixSim overlap $\omega_{j\ell} = \Pr_{f_j}(\pi_\ell f_\ell > \pi_j f_j) + \Pr_{f_\ell}(\pi_j f_j > \pi_\ell f_\ell)$, averaged over pairs into $\bar{\omega}$ [23, 42]
  - exact route for $G < 4$ (`MixSim::overlap()`); for $G \geq 4$, our Monte Carlo route draws n points per component from a scrambled Sobol sequence mapped through Φ^{-1} and the Cholesky factor of $\boldsymbol{\Sigma}_j$, and counts Bayes-rule misassignments (distance vignette)
  - Monte Carlo error is close to the binomial $\sqrt{\omega(1-\omega)/n}$: standard deviations of 0.0008 to 0.0032 with $n = 4000$ over the three fig03 targets, and agreement with the exact route to within one standard deviation on a three-gene block
  - for two equally weighted components, $\omega = 1 - \mathrm{TV}(f_j, f_\ell)$; the closed-form Hellinger distance brackets it, $1 - H\sqrt{2 - H^2} \leq \omega \leq 1 - H^2$, and is kept as a cheap descriptor
- descriptors reported with every scenario, with no composite score

Confirm these Monte Carlo errors with the script seed and report them in the vignette.

  - composition: $H^\star$, effective number of types, $\|\boldsymbol{p}\|_2^2$
  - mean geometry: pairwise cosines, Euclidean gaps, condition number of $\boldsymbol{\mu}$
  - conditioning of $\boldsymbol{\Sigma}(\boldsymbol{p})$
  - tangent Fisher information and the share carried by the covariance block [43]
  - edge density of each precision graph
  - separation: MixSim overlap, mean pairwise Hellinger distance, affine-invariant Riemannian distance
- scores and their pitfalls: Appendix F

3.7 From a single-cell reference to DeCovarT inputs

3.7.1 Suppinger gastruloids

- source: 10x Chromium 3′ UMI profiles of murine gastruloids at 48, 72 and 96 hours, with annotated cell states [1]
- one technical scRNA-seq experiment: cells are draws from one specimen, not biological replicates [44, 45]
- models are fitted separately at each time point; types with fewer than 20 cells are left out

Extract batches, viability and matched bulk profiles, if any, and state which analyses use them.

3.7.2 Which genes enter the signature

- aim: genes that separate close types and that have a non-degenerate distribution in every type
- presence gate, fixed before ranking: detection rate $\pi_{gj} \geq 0.05$ and within-type variance $v_{gj} > 0$ in every type; a constant gene makes $\boldsymbol{\Sigma}_j$ singular
- main effect of cell type, tested against the technical null
  - $M_0 : \log \mu_{cg} = \beta_{0g} + \log L_c$, $y_{cg} \sim \mathrm{NB}(\mu_{cg}, \theta_g)$, with library size L_c [46]; M_1 adds the type effect, a joint test on $J - 1$ degrees of freedom [47]
  - likelihood-ratio statistic $D_g(M_0) - D_g(M_1)$; the Wald test is avoided because an all-zero type sends $\hat{\beta}_{gj} \to -\infty$ (Hauck–Donner effect) [39]
  - the between-type sum of squares of Pearson residuals under M_0 is, up to weight heterogeneity, the Rao score statistic of this test; divided by the within-type dispersion it is a quasi-likelihood F test [48, 49]
  - overdispersed genes are kept: their within-type scatter is the covariance DeCovarT models
- calibration and stability
  - the nominal F tail is optimistic on clipped, discrete residuals [50]; the null is calibrated by permuting cell-type labels, with Freedman–Lane residual permutation when covariates are present [51]

  - a bootstrap of cells within type measures how often a gene stays in the panel [52]; it does not generate a null
- ranking: with thousands of cells nearly every expressed gene rejects the null, so the test is a gate and genes are ranked by effect size (between-type share of variance, deviance explained, or the smallest pairwise Hellinger distance between type-specific laws)
- the p-values calibrate a screen inside one specimen, not population-level differential expression
- details: feature-selection chapter of the reproducibility book

3.7.3 Cell-type-specific covariance and precision

- target: one precision $\mathbf{\Omega}_{j,t}$ per type and time point; with one specimen, the estimand is the within-type cell-to-cell covariance
- five estimators, compared in the gastruloid ablation
  - partial correlations by inversion with Benjamini–Hochberg edges and a positive-definite refit on the support [53]; valid only when $n_j > G$
  - graphical lasso [30] on the correlation matrix, with an unpenalised diagonal and a per-type extended BIC [54], then rescaled to the data
  - de-sparsified graphical lasso on nonparanormal scores with global false discovery rate control [55–58]
  - Poisson–log-normal network on raw UMI counts with a library-size offset [28]
  - dense shrinkage of the correlations with the variances kept [59], judged by held-out Gaussian log-likelihood
- sparsity is secondary for DeCovarT: the likelihood needs $\mathbf{\Sigma}_j$, and $\sum_j p_j^2 \mathbf{\Sigma}_j$ has no sparse precision even when every $\mathbf{\Omega}_j$ is sparse
- empty graphs are a detection limit: above $\max_{g \neq g'} |R_{gg'}|$ the graphical lasso returns a diagonal precision, and recovery needs $n \asymp d^2 \log G$ for maximum degree d [60, 61]; the dense shrinkage estimate is the fallback plug-in
- review of the estimators and their conditions: Appendix A

3.7.4 Tailoring DeCovarT to a single-cell reference

- setting: one UMI reference from one donor, one bulk sample at a time, no external covariate, no gene-wise scale estimated from other bulks, and no split of size factors into technical and biological parts
- the released covariance $\sum_j p_{ji}^2 \mathbf{\Sigma}_j$ is exact for one random population profile per type (sorted references); a single-cell reference supplies a cell-to-cell covariance, and the bulk is a physical sum of cells
- reference moments, computed on raw UMI counts without library-size normalisation: $\tilde{\boldsymbol{\mu}}_{\cdot j} = C_j^{-1} \sum_{c \in j} \boldsymbol{x}_{\cdot c}$, so that $S_j = \mathbf{1}^\top \tilde{\boldsymbol{\mu}}_{\cdot j}$ is the mean transcriptome size of a type-j cell, and $\mathbf{\Sigma}_{W,j}$, the within-type cell-to-cell covariance estimated after removing the Poisson diagonal and the technical size-factor term [62, 63], then regularised (Section 3.7.3)

- bulk: $\boldsymbol{y}_{\cdot i} = a_i \sum_j \boldsymbol{T}_{ji}$, with $m_{ji} = n_i p_{ji}$ independent cells per type; the bulk expression per cell on the reference scale is $\tilde{\boldsymbol{y}}_{\cdot i} = \boldsymbol{y}_{\cdot i}/(a_i n_i)$, that is $\boldsymbol{y}_{\cdot i}/n_i$ when the two platforms share their scale ($a_i = 1$)
- recommended model (final choice), exact for independent Gaussian cells and a central-limit approximation for counts, since n_i runs from thousands to millions

$$\boldsymbol{y}_{\cdot i} \mid \boldsymbol{p}_{\cdot i} \sim \mathcal{N}_G\Big(a_i n_i\, \tilde{\boldsymbol{\mu}} \boldsymbol{p}_{\cdot i},\ a_i^2 n_i \sum_j p_{ji} \boldsymbol{\Sigma}_{W,j}\Big), \qquad \boldsymbol{p}_{\cdot i} = \operatorname{softmax}(\mathbf{V}\boldsymbol{\rho}_i), \tag{5}$$

equivalently $\tilde{\boldsymbol{y}}_{\cdot i} \sim \mathcal{N}_G\big(\tilde{\boldsymbol{\mu}} \boldsymbol{p}_{\cdot i},\ n_i^{-1} \boldsymbol{S}_W(\boldsymbol{p}_{\cdot i})\big)$

- estimand: $\tilde{\boldsymbol{\mu}}$ keeps the cell sizes S_j, so $\boldsymbol{p}_{\cdot i}$ are cell fractions directly; transcript fractions follow as $q_{ji} = p_{ji} S_j / \sum_\ell p_{\ell i} S_\ell$, with no post-fit conversion [5, 64]; only the ratios S_j/S_ℓ matter
- covariance weights (our remark): an earlier version of this model weighted the cell-to-cell covariance by transcript fractions q_{ji}; independent cells add with their counts $m_{ji} = n_i p_{ji}$, so the linear layer is weighted by $\boldsymbol{p}_{\cdot i}$
- scales: a_i and n_i enter only through $c_i = a_i n_i$ in the mean and $\phi_i = a_i c_i = a_i^2 n_i$ in the covariance, two free positive scalars; n_i reads as a number of cells only when a_i is known (same platform, or paired samples)
- computing $\tilde{\boldsymbol{y}}_{\cdot i}$: since $\mathbf{1}^\top \mathbb{E}\boldsymbol{y}_{\cdot i} = c_i \sum_j p_{ji} S_j$, the library size gives the moment estimator $\hat{c}_i = \mathbf{1}^\top \boldsymbol{y}_{\cdot i} / \sum_j \hat{p}_{ji} S_j$, and $\hat{\tilde{\boldsymbol{y}}}_{\cdot i} = \boldsymbol{y}_{\cdot i}/\hat{c}_i$; the likelihood profile of Section 3.7.5 replaces it by a generalised least-squares estimate
- no extra bulk noise: a Poisson or negative-binomial measurement term $\phi\, \operatorname{diag}(\boldsymbol{m}_i)$ is left out; with one bulk sample and no technical replicate it cannot be separated from the scale ϕ_i and from the cell-type variability without further assumptions
- one donor cannot identify the between-donor layer $\boldsymbol{S}_B(\boldsymbol{p}_{\cdot i})$ [44]; with $R \geq 2$ donors it is added as $a_i^2 n_i^2 \boldsymbol{S}_B(\boldsymbol{p}_{\cdot i})$ (Appendix D.3)
- link with Poisson measurement models
  - *RCTD* writes $y_{gi} \sim \text{Poisson}(N_i \sum_j w_{ji} \mu_{jg} e^{\gamma_g})$ with log-normal overdispersion, per-UMI reference profiles μ_{jg}, a platform ratio $\tilde{b}_g = e^{\gamma_g}$ and $\boldsymbol{w}_{\cdot i} = \boldsymbol{p}_{\cdot i} e^{\alpha_i}$ left unconstrained [12]; Sarkar and Stephens write a UMI count as $\text{Poisson}(x_+ b_g \lambda_g)$ given the total x_+ and the capture b_g [62]
  - (5) has the same multiplicative mean, but the rate is carried by cell types: an average type-j cell contributes $S_j \boldsymbol{\theta}_{\cdot j}$, multiplied by the number of cells and by the sample scale, instead of a free rate per gene
  - its variance comes from the cell-to-cell covariance, not from the Poisson law that equates variance and mean, which is why it carries gene–gene dependence
  - the scale (n in *RNA-Sieve*, $N_i e^{\alpha_i}$ in *RCTD*, c_i here) is not, in general, the number of cells: that reading needs the same sequencing platform, or paired samples with similar biological and technical size factors
- other alignment strategies and why they were not chosen: Appendix D.3

3.7.5 Estimation with Newton–Raphson on the ILR chart

- parameters $\boldsymbol{\vartheta} = (\boldsymbol{\rho}, \log c, \log \phi)$ for one bulk sample, with $c = an$ and $\phi = a^2 n$; with $\boldsymbol{W} = \boldsymbol{S}_W(\boldsymbol{p})^{-1}$ and $\boldsymbol{r} = \boldsymbol{y} - c\tilde{\boldsymbol{\mu}}\boldsymbol{p}$,

$$\ell(\boldsymbol{\vartheta}) = -\frac{G}{2}\log\phi - \frac{1}{2}\log\det \boldsymbol{S}_W(\boldsymbol{p}) - \frac{1}{2\phi}\,\boldsymbol{r}^\top \boldsymbol{W}\boldsymbol{r} \tag{6}$$

- closed-form profiles for fixed $\boldsymbol{p}$: the generalised least-squares scale $\hat{c}(\boldsymbol{p}) = (\tilde{\boldsymbol{\mu}}\boldsymbol{p})^\top \boldsymbol{W}\boldsymbol{y}/(\tilde{\boldsymbol{\mu}}\boldsymbol{p})^\top \boldsymbol{W}\tilde{\boldsymbol{\mu}}\boldsymbol{p}$ and $\hat{\phi}(\boldsymbol{p}) = Q(\boldsymbol{p})/G$, with Q the minimised Mahalanobis residual $\hat{\boldsymbol{r}}^\top \boldsymbol{W}\hat{\boldsymbol{r}}$; the profile log-likelihood is $\ell^*(\boldsymbol{\rho}) = -\frac{G}{2}\log Q(\boldsymbol{\psi}(\boldsymbol{\rho})) - \frac{1}{2}\log\det \boldsymbol{S}_W(\boldsymbol{\psi}(\boldsymbol{\rho}))$, a function of $J-1$ coordinates
- profile score, by the envelope theorem, $\partial\ell^*/\partial p_j = (G/Q)\{\hat{c}\,\tilde{\boldsymbol{\mu}}_{\cdot j}^\top \boldsymbol{W}\hat{\boldsymbol{r}} + \frac{1}{2}\hat{\boldsymbol{r}}^\top \boldsymbol{W}\boldsymbol{\Sigma}_{W,j}\boldsymbol{W}\hat{\boldsymbol{r}}\} - \frac{1}{2}\operatorname{tr}(\boldsymbol{W}\boldsymbol{\Sigma}_{W,j})$, mapped to ILR coordinates by $\mathbf{J}_\psi^\top$
- Newton–Raphson on the full $\boldsymbol{\vartheta}$: score and Hessian of a Gaussian with parametrised mean and covariance (general score and Hessian); the mean $c\tilde{\boldsymbol{\mu}}\boldsymbol{p}$ and the covariance $\phi\boldsymbol{S}_W(\boldsymbol{p})$ are linear in $\boldsymbol{p}$, so the only second derivatives in $\boldsymbol{p}$ come from the curvature of the chart (D7)
- expected information: $I_{c\phi} = 0$, $I_{\phi\phi} = G/(2\phi^2)$, $I_{p_j\phi} = \operatorname{tr}(\boldsymbol{W}\boldsymbol{\Sigma}_{W,j})/(2\phi) \neq 0$ and $I_{p_j c} = c\,\tilde{\boldsymbol{\mu}}_{\cdot j}^\top \boldsymbol{W}\tilde{\boldsymbol{\mu}}\boldsymbol{p}/\phi \neq 0$; the covariance of $\hat{\boldsymbol{\rho}}$ is the inverse of the Schur complement $I_{\rho\rho} - I_{\rho\eta}I_{\eta\eta}^{-1}I_{\eta\rho}$, $\boldsymbol{\eta} = (c, \phi)$, so estimating the scales widens the intervals automatically
- start from a non-negative least-squares fit $\boldsymbol{y} \approx \tilde{\boldsymbol{\mu}}\boldsymbol{w}$, with $\boldsymbol{p}^{(0)} = \boldsymbol{w}/\mathbf{1}^\top\boldsymbol{w}$ floored away from zero and $c^{(0)} = \mathbf{1}^\top\boldsymbol{w}$, plus Dirichlet multistarts because ℓ^* is not concave; stop on the norm of the ILR score, which does not depend on the Helmert basis
- joint versus plug-in nuisance scales (plug-in strategy)
  - with one bulk sample and no external data there is no genuine plug-in: a two-step moment estimate of c and ϕ from the same bulk is possible, but the variance of $\hat{\boldsymbol{p}}$ then needs the correction $I_{pp}^{-1} + I_{pp}^{-1}I_{p\eta}\boldsymbol{V}_\eta I_{\eta p}I_{pp}^{-1}$ for an estimate $\hat{\boldsymbol{\eta}}$ of covariance $\boldsymbol{V}_\eta$; the joint fit accounts for it through the Schur complement
  - external data would make one scale a true plug-in: cell counts per sample give n_i, spike-ins or paired bulk and single-cell profiles give a_i or $\tilde{\boldsymbol{b}}$, and then $\boldsymbol{p}_{\cdot i}$ are cell fractions free of technical scale
- limitations (our assessment)
  - $N = 1$: the G genes are not independent replicates, so asymptotics in G need weak dependence between genes, as assumed by *MEAD* [15]
  - one donor gives no between-donor layer: ϕ inflates the covariance globally but cannot correct its shape, so the intervals are expected to under-cover; the sandwich covariance or the reference bootstrap of cells (`reference_bootstrap_decovart(method = "cells")`) is the safeguard
  - dropping the bulk Poisson diagonal assumes a deep bulk library; $\boldsymbol{\Sigma}_{W,j}$ must be the denoised biological covariance, otherwise the reference sampling noise is counted n_i times

– the determinant term of (6) adds information only if the variance function is right; otherwise the quasi-score of Section 4.2.4 is the safer estimator

Implement (5) in `fit_decovart()` (covariance assembler linear in $\boldsymbol{p}$, profiled c and ϕ, Schur-complement Wald intervals) and validate it in an ADEMP simulation study [25] before any claim rests on it.

3.7.6 Estimand: transcript fraction versus cell fraction

- under (2), $\boldsymbol{p}$ is the relative transcriptomic contribution of each type
- it equals the cell fraction only if capture efficiency and RNA content per cell are shared across types [64]; (5) estimates cell fractions directly because $\tilde{\boldsymbol{\mu}}$ keeps the cell sizes S_j
- absolute cell counts are not targeted: capture efficiency is not observed
- library-size closure (TPM, RPKM) is avoided: dividing by the total induces negative gene–gene correlation, including for genes that were independent

3.8 Performance metrics

- primary accuracy scores on the simplex: total variation $\frac{1}{2}\|\hat{\boldsymbol{p}}-\boldsymbol{p}^\star\|_1$, RMSE and the maximum absolute error; all are finite with zeros, but abundant types dominate them
- compositional scores (Aitchison distance) measure fold-change error and weigh rare types fairly, but are undefined at zero; they are reported only on interior estimates, with the fraction of zeros [8]
- Monte Carlo summaries: bias, empirical standard deviation, the ratio of model standard error to empirical standard deviation, coverage with its Wilson interval [65], and mean interval width
- optimisation: numerical and theoretical convergence, and the projected-score residual
- formulas, pitfalls and the recommended table: Appendix F

4 Discussion and Perspectives

4.1 DeCovarT's core innovations

- cell-type composition is a candidate prognostic marker of gastruloid fate
- single-cell RNA-seq resolves abnormal cell states, at higher resolution than bulk, but cost and dissociation limit it to few structures and one time point per structure
- DeCovarT collects, in one likelihood, ideas from algorithms that model within-type variability
  - *MuSiC*: one scale aligns the bulk average with the single-cell average, for library size and capture efficiency across platforms [5]

  - *RNA-Sieve*: Gaussian cell-type profiles justified by a central-limit argument on the moments, with independent genes; DeCovarT drops gene independence. RNA-Sieve reports Wald regions from the inverse Fisher information of its gene-wise CLT likelihood, and a Godambe sandwich when protocol shift takes the truth outside the model family [10]. DeCovarT's analogue of the first is (4); it is not calibrated on a simplex face, where the chi-bar-square likelihood-ratio test and a restricted parametric bootstrap are required.

    RNA-Sieve's noisy-M term is the likelihood analogue of resampling the reference; DeCovarT currently keeps $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_j$ plug-in and propagates that uncertainty by a donor- or cell-level bootstrap. [10]
  - *DECALS*: composition and cell-type covariances estimated together, so within-type and cross-sample variability both enter the fit [14]
  - *MEAD*: an errors-in-variables Gaussian model with confidence intervals for the proportions [15]
- benchmarks that cross algorithms with normalisation choices use raw counts as the proxy of mRNA abundance; library-size normalisations turn the problem into a composition on the gene simplex
- the scaling assumptions of Figure 1 are unlikely to hold; absolute mRNA content per type then needs an external reference or an orthogonal assay
- estimand: share of transcriptomic content per type, not the cell-count composition

4.2 Improve DeCovarT

4.2.1 Extended benchmark on true aliquots

Real-data evaluation should start from matched flow-cytometry / bulk RNA-seq panels such as the Kassandra benchmark [66], which already compares EPIC [64], CIBERSORT [2], CIBERSORTx [3], quanTIseq [67] and ABIS [68].

4.2.2 Evaluate the impact of normalisation techniques

A systematic study of the impact of transformation and normalisation techniques, such as CPM or a $\log_2$ transform, on the covariance estimates and on the performance of DeCovarT is left to a later benchmark; the closure induced by library-size normalisation is discussed in Section 1.2.

Briefly report the content of [69] on the impact of normalisation techniques

.

4.2.3 A lineage-aware framework for closely related cell types

- a lineage tree can constrain the composition, either by hard parent–child equality [5, 70–72] or by a soft mismatch penalty [73]
- continuous states can replace discrete types through archetypes [74, 75], which suits the potency gradient of gastruloids

4.2.4 A generalised linear model for complex designs

DeCovarT already has the structure of a generalised linear model (GLM). It models the mean of the response through an inverse link on the parameter, $\boldsymbol{m}(\boldsymbol{\rho}) = c\,\tilde{\boldsymbol{\mu}}\boldsymbol{\psi}(\boldsymbol{\rho})$ with $\boldsymbol{\psi}$ the softmax map of the ILR chart, instead of transforming the response, which would change the estimand and break the additive mixture [47]. Its variance function $\boldsymbol{V}(\boldsymbol{\rho}) = \phi\,\boldsymbol{S}_W(\boldsymbol{\psi}(\boldsymbol{\rho}))$ depends on the same parameter, up to the dispersion ϕ of (5); with $R \geq 2$ reference donors it gains the quadratic between-donor layer $\boldsymbol{S}_B$. Transformation-based methods such as *DeOPUS* fix the gene weights in advance and have no likelihood [76].

Quasi-likelihood version.

Only the mean–variance relationship matters for the composition, so the model can be stated by its first two moments, without a distributional form (GLM view).

- specification: $\mathbb{E}(\boldsymbol{y} \mid \boldsymbol{\rho}) = \boldsymbol{m}(\boldsymbol{\rho})$ and $\mathrm{Cov}(\boldsymbol{y} \mid \boldsymbol{\rho}) = \phi\boldsymbol{V}_0(\boldsymbol{\rho})$, with $\boldsymbol{V}_0 = \boldsymbol{S}_W(\boldsymbol{\psi}(\boldsymbol{\rho}))$ known up to ϕ
- quasi-score $\boldsymbol{D}^\top \boldsymbol{V}_0^{-1}\{\boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\rho})\} = \boldsymbol{0}$, with $\boldsymbol{D} = \partial\boldsymbol{m}/\partial\boldsymbol{\rho}^\top = c\,\tilde{\boldsymbol{\mu}}\mathbf{J}_\psi$; it is unbiased whenever the mean is right, whatever the law of the residuals
- information $\phi^{-1}\boldsymbol{D}^\top \boldsymbol{V}_0^{-1}\boldsymbol{D}$, the mean part of the Fisher information; the Gaussian likelihood adds the covariance part, which is informative only if $\boldsymbol{V}_0$ is right
- dispersion from the Pearson statistic, $\hat{\phi} = \{\boldsymbol{y} - \boldsymbol{m}(\hat{\boldsymbol{\rho}})\}^\top \boldsymbol{V}_0^{-1}\{\boldsymbol{y} - \boldsymbol{m}(\hat{\boldsymbol{\rho}})\}/(G - J)$, and a sandwich covariance when $\boldsymbol{V}_0$ is misspecified
- the iteration is Fisher scoring, with the same ILR chart and Newton machinery as Section 3.7.5

Several bulk samples.

- current scope: independent bulk samples and one single-cell reference donor; each composition is a free parameter
- composition model across samples: $\boldsymbol{\rho}_i = \boldsymbol{B}\boldsymbol{z}_i + \boldsymbol{u}_i$, with $\boldsymbol{z}_i$ a design row (time point, condition, batch), $\boldsymbol{B}$ a $(J-1) \times K$ coefficient matrix and $\boldsymbol{u}_i$ a sample effect; each sample keeps the likelihood (3), and the samples are tied through $\boldsymbol{B}$
- benefits
  - shared technical scales (a_i, n_i, gene-wise ratios $\tilde{\boldsymbol{b}}$) estimated jointly over samples, and therefore more stably than from one bulk sample
  - contrast matrices on $\boldsymbol{B}$ test differential abundance of cell types between conditions, with likelihood-ratio or Wald tests that use the covariance of $\hat{\boldsymbol{\rho}}_i$
- precedent: *MEAD* regresses $\mathbb{E}\boldsymbol{p}_{\cdot i} = h(\boldsymbol{\beta}_0 + \boldsymbol{A}^\top \boldsymbol{f}_i)$ by estimating equations with a sandwich variance; plugging in estimated proportions is valid when $N = o(G)$, and the reported coverage of group differences falls from 0.93 with 86 samples to 0.81 with 1000 [15]
- caution: regressing ILR coordinates linearly on $\boldsymbol{z}_i$ targets $\mathbb{E}(\mathrm{ilr}\,\boldsymbol{p})$, not $\mathrm{ilr}\,\mathbb{E}(\boldsymbol{p})$, so the two parametrisations answer different questions

- sample-level covariates cannot be appended as extra columns of $\boldsymbol{\mu}$; they may instead modulate the cell-state moments $\boldsymbol{\mu}_{\cdot j}(\boldsymbol{z}_i)$, as *MuSiC2* does by removing condition-specific genes [77], or the log-ratio coordinates as above (GLM view)

Implement the design-based fit $\boldsymbol{\rho}_i = \boldsymbol{B}\boldsymbol{z}_i + \boldsymbol{u}_i$ and evaluate the coverage of contrasts on $\boldsymbol{B}$ in simulation, as a function of N/G.

4.2.5 Robust regression: down-weighting inconsistent genes

- DeCovarT as generalised least squares (GLS)
  - for a fixed weight matrix, the Mahalanobis term of (3) is the GLS criterion (Theorem 2); weighted least squares, as in *DWLS*, is its diagonal case [6]
  - *DECALS* solves a constrained GLS with a covariance estimated from the bulk cohort [14]; DeCovarT uses a known, plug-in covariance whose weight $\boldsymbol{\Sigma}(\boldsymbol{p})^{-1}$ moves with the composition, plus a log-determinant term
  - DeCovarT is therefore a GLS with a parameter-dependent variance function, close to iteratively reweighted least squares in a quasi-likelihood model
  - a simpler competitor freezes the weight at the barycentre, $\boldsymbol{\Sigma} = J^{-2}\sum_j \boldsymbol{\Sigma}_j$; it ignores the cell-type-specific networks, and *cellGeometry* estimates such a gene-level covariance from cross-sample variances [78]
- DeCovarT is not robust in the statistical sense: the loss is quadratic in the whitened residual $\boldsymbol{e} = \boldsymbol{L}^{-1}(\boldsymbol{y} - \boldsymbol{\mu}\boldsymbol{p})$, so one gene with a platform shift, or a gene expressed by a type missing from the reference, has unbounded influence on $\hat{\boldsymbol{p}}$; fixed gene weights rescale that influence without bounding it
- robust rewrites, in increasing departure from the current solver
  1. fixed gene weights folded into the convolution, $\boldsymbol{y}^\star = \boldsymbol{W}^{1/2}\boldsymbol{y}$, $\boldsymbol{\mu}^\star = \boldsymbol{W}^{1/2}\boldsymbol{\mu}$, $\boldsymbol{\Sigma}_j^\star = \boldsymbol{W}^{1/2}\boldsymbol{\Sigma}_j\boldsymbol{W}^{1/2}$, as *BayesPrism* does with *DWLS* weights [79]; the solver is unchanged
  2. M-estimation on gene-wise standardised residuals $e_g = r_g/\{\boldsymbol{\Sigma}(\boldsymbol{p})\}_{gg}^{1/2}$: replace $\frac{1}{2}\sum_g e_g^2$ by $\sum_g \varrho_c(e_g)$ with a Huber or bisquare loss [80, 81], solved by iteratively reweighted least squares with weights $w_g = \varrho_c'(e_g)/e_g$, and report the sandwich variance $\boldsymbol{H}^{-1}\boldsymbol{K}\boldsymbol{H}^{-1}$ (Appendix D.5.3)
  3. trimmed likelihood: drop the genes with the largest contributions to (3) and refit, as least trimmed squares does for observations [82]
  4. robust reference moments: estimate $\boldsymbol{\Sigma}_j$ from cells with a high-breakdown estimator such as the minimum covariance determinant [83], so outlier cells do not distort the plug-in
- caution: the Cholesky whitening mixes genes, so down-weighting one whitened coordinate does not down-weight one gene; gene-level weights must act on gene-wise standardised residuals

4.2.6 Other observation laws allow correlation between genes

- *RNA-Sieve*, *MEAD* and *DECALS* support a multivariate Gaussian model for the genes, and only asymptotically; none states a lower bound on the number of single-cell profiles per type
- with few profiles, a logit-normal law can give better-calibrated intervals [84]
- multivariate counts rarely have a closed form (negative-binomial or log-normal margins), so a convolution under those laws needs numerical integration or Monte Carlo
- alternative laws already used for deconvolution
  - discrete counts: *ISOpureR*, *BayesPrism* (multinomial) [19, 85]; *DeconV* (Poisson) [13]; Poisson–log-normal and zero-inflated extensions [86, 87]; *RCTD* [12]
  - continuous, gene-wise: *BLADE* [17]; *bMIND* [88]
- *CSNet* reverses (2): $\boldsymbol{p}$ treated as known, type-level means and a sparse correlation estimated by least squares [89]; its output is a co-expression graph, and a graphical lasso on $\hat{\boldsymbol{\Sigma}}_j$ gives the precision DeCovarT can plug in

4.2.7 Deconvolution in a latent space: factor models and foundation-model embeddings

DeCovarT deconvolves in the original gene space. An alternative projects the bulk profiles and the reference into a lower-dimensional latent space and solves the mixture there.

- linear latent factor models keep the mixture linear, because a linear projection maps $\boldsymbol{y} = \boldsymbol{\mu}\boldsymbol{p}$ to $\boldsymbol{L}\boldsymbol{y} = (\boldsymbol{L}\boldsymbol{\mu})\boldsymbol{p}$
  - non-negative matrix factorisation writes the bulk matrix as a product of non-negative profiles and proportions, reference-free or guided by markers [90, 91]; NMF is ill-posed, and *GSNMF+* restores the solvability conditions by augmenting the bulk with pseudo-bulks simulated from single-cell data, then corrects the remaining rescaling of the solution [92]
  - convex analysis of mixtures uses the geometry of the problem: bulk samples lie in the convex hull of the subtype profiles, the vertices identify marker genes, and minimum description length sets the number of subtypes [93]
  - *swCAM* adds sample-wise deviations $\Delta\boldsymbol{S}_i$ to the subtype profiles, penalised by a nuclear norm and solved by ADMM, to recover subtype-specific expression in each sample; recovery is good for abundant subtypes and poor for rare ones (estimate–truth correlation from 0.20 to 0.71 across abundance quartiles in its realistic simulation) [94]
- non-linear embeddings learn the latent space
  - deep networks trained on simulated pseudo-bulks predict proportions directly [95]
  - *DECONVersation* embeds each bulk sample, treated as a pseudo-cell, and each cell-type signature with single-cell foundation models (Geneformer,

Cell2Sentence, scGPT, CellHermes), then runs non-negative least squares on the embeddings [96]; on six bulk datasets with ground truth it matched *DWLS* and outperformed *MuSiC* and *BayesPrism*, and its advantage grew with the technical dissimilarity between reference and bulk

- advantages of a latent space
  - robustness: embeddings trained on large and heterogeneous corpora are less sensitive to batch and platform effects, and rank-based tokenisation acts as a normalisation
  - no marker-selection step, whose choice changes the estimates; the signature is also better conditioned in embedding space than on raw genes
  - gene interactions: mean-only methods ignore them, and DeCovarT captures only linear dependence through a Gaussian graphical model; deep models can encode non-linear, higher-order interactions between genes
- drawbacks
  - interpretation: latent coordinates carry no direct biological meaning, and attention-based marker enrichment is an indirect check
  - the linear relationship between bulk and single-cell expression breaks: for a non-linear map f, $f(\sum_j p_j \boldsymbol{x}_{\cdot j}) \neq \sum_j p_j f(\boldsymbol{x}_{\cdot j})$, so the proportions are no longer the parameters of a generative model; *DECONVersation* checks additivity only by a cosine similarity that anisotropy inflates, and bulk samples are out of distribution for models trained on single cells
  - inference: without a likelihood there is no finite-sample or asymptotic interval for $\boldsymbol{p}$, no boundary test for an absent type, and no marginal effect of explanatory covariates (Section 4.2.4); resampling remains, at a high computational cost
  - cost and variability: fine-tuning needs a GPU, and results change with the chosen model
- a bridge we suggest: a fixed linear projection $\boldsymbol{L}$, such as factor loadings, keeps DeCovarT exact on $\boldsymbol{y}^\star = \boldsymbol{L}\boldsymbol{y}$, $\boldsymbol{\mu}^\star = \boldsymbol{L}\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_j^\star = \boldsymbol{L}\boldsymbol{\Sigma}_j\boldsymbol{L}^\top$, so a low-dimensional covariance-driven fit keeps its likelihood and intervals; non-linear embeddings are then better used for gene or feature selection than as the space of the likelihood

4.2.8 A Bayesian framework: the Wiener filter

- keeping the profiles as latent variables, with a diagonal residual $\boldsymbol{D}$, $\boldsymbol{y}_{\cdot i} \mid \boldsymbol{p}_{\cdot i}, \{\boldsymbol{x}_{\cdot j,i}\} \sim \mathcal{N}_G(\sum_j p_{ji}\boldsymbol{x}_{\cdot j,i}, \boldsymbol{D})$, returns the DeCovarT likelihood after integration; $\boldsymbol{D} \neq \boldsymbol{0}$ is needed for a non-degenerate posterior
- given $\boldsymbol{p}_{\cdot i}$ the maximum a posteriori estimate, the best linear unbiased predictor and the Wiener filter coincide: $\hat{\boldsymbol{x}}_{\cdot j,i} = \boldsymbol{\mu}_{\cdot j} + p_{ji}\boldsymbol{\Sigma}_j\{\boldsymbol{\Sigma}(\boldsymbol{p}_{\cdot i}) + \boldsymbol{D}\}^{-1}(\boldsymbol{y}_{\cdot i} - \boldsymbol{\mu}\boldsymbol{p}_{\cdot i})$; the cell-type covariance then shapes the purified profiles as well as $\hat{\boldsymbol{p}}$
- the posterior covariance gives per-sample uncertainty on cell-type-specific profiles, which *bMIND* obtains with a gene-wise single-cell prior [88]
- the latent form yields an EM algorithm [97] whose iterations increase the marginal likelihood, and a mean-field variant that keeps the sparse precisions

- priors on the composition: Dirichlet, concave in ILR coordinates for every $\alpha_j > 0$, or logistic-normal; either keeps the posterior proper where Wald intervals degenerate
- routes to the full posterior: Laplace approximation with the analytic Hessian, Hamiltonian Monte Carlo on the closed-form marginal likelihood, or stochastic variational inference
- open problems: nothing is implemented in `fit_decovart()`; EM stalls as $\boldsymbol{D} \to \mathbf{0}$; plugging in $\hat{\boldsymbol{p}}$ ignores its uncertainty (Wiener-filter vignette)

4.2.9 A dynamic and time-resolved perspective

- a few time points (at least three) are needed to recover dynamic interactions between genes
- without prior knowledge or perturbation data, temporal ordering is the main route to directed, mechanistic interactions, and hence to sparser and more contextualised networks
- time-resolved compositions can carry a random-walk prior on log-ratio coordinates [98]

4.2.10 Alternative strategies for cell-type-specific covariances

Besides the mean of each cell type, DeCovarT needs $\boldsymbol{\Sigma}_j$ or $\boldsymbol{\Omega}_j$. Gene–gene dependence estimated from single cells remains an open problem. Specifically, inferring mechanistic and directed regulatory edges typically requires additional data modalities, such as chromatin accessibility (ATAC-seq) or transcription-factor binding information (TF-seq) information [99, 100].

- the fitted precision is undirected, while most regulatory interactions are directed
  - orient edges by pairing observations with interventions
  - or add chromatin accessibility or transcription-factor binding [99, 100]
- benchmarks show that inference can fail in common regimes even with multi-omic inputs [101]
- open question: whether a causal graph improves $\hat{\boldsymbol{p}}$, or a consistent covariance is enough
- current fits use the variability inside one single-cell biological replicate; several independent replicates would separate that within-type term from cross-sample variability, for example with a two-level random effect, as *DECALS* already does [14]

4.2.11 Roadmap of other extensions

Each extension keeps the convolution (2) and changes one part of it; none is implemented (roadmap vignette).

- composition structure: sample covariates in a composition model on the log-ratio chart [77]

- reference completeness: one unknown component absorbing an unrepresented population [64, 102, 103]
- observation space: isoforms [104], stacked multi-omics with a block covariance [105, 106], and spatial spots whose compositions depend on their neighbours [12]
- estimation: ensembles over references [107], gene weights folded into the convolution as $\boldsymbol{W}^{1/2}\boldsymbol{y}$, $\boldsymbol{W}^{1/2}\boldsymbol{\mu}$ and $\boldsymbol{W}^{1/2}\boldsymbol{\Sigma}_j\boldsymbol{W}^{1/2}$ with the solver unchanged [6, 108], and robust losses [109]

Code availability. DeCovarT is an R package [110], available on GitHub with its vignettes and the scripts that generate every figure.

- numerical optimisation: Newton–Raphson on the ILR chart with `stats::nlminb` (default); Marquardt–Levenberg with the relative-distance-to-maximum criterion from `marqLevAlg` [33, 111]; L-BFGS-B from `stats::optim` [112]; matrix algebra and Cholesky factors with `Matrix` and `MASS` [113, 114]; analytic derivatives checked with `numDeriv` [31]; mean-only baselines with `limSolve` [115]
- simulation: multivariate Gaussian draws with `mvtnorm` [116]; random graphs with `igraph` [117]; exact overlap with `MixSim` [118] and Sobol sequences for the Monte Carlo overlap with `qrng` [119]; graphical lasso with `huge` [120]; parallel replicates with `furrr` and independent L'Ecuyer streams [121]
- visualisation: `ggplot2` [122], `ggdist` [123], `ComplexHeatmap` [124] and `cowplot` [125]
- random seeds are fixed in every script, and session information is saved with each output

State the R version, the computing infrastructure
and the package versions used for the final figures.

Author contributions. B.C. conceived the method, implemented the `DeCovarT` package, designed the simulation studies, and wrote the manuscript. M.P. contributed equally to the methodological development and manuscript drafting. M.A.C. contributed to the benchmarking of gene-regulatory-network inference techniques used in the synthetic scenarios. E.V. suggested the ablation study and the GRNBoost2-based gene selection. G.N., E.B. and A.B. jointly supervised the work and revised the manuscript. All authors approved the final version.

Conflicts of interest. The authors declare no conflict of interest.

Acknowledgements. We thank Boris Hejblum for advice on computational performance and numerical precision, notably the use of the log-sum-exp trick for evaluating the additive logistic (softmax) map and its derivatives. We thank Kalidou Ba for suggestions on pre-processing and normalisation of bulk and single-cell omics inputs to deconvolution algorithms. We thank Annesh Pal for ideas on more realistic synthetic scenarios for DeCovarT, in particular the use of random graph models to generate richer topological structures for gene-regulatory networks. Elias Ventre, whose work covers stochastic models of gene expression, gene-regulatory-network inference and

trajectory reconstruction from single-cell data, suggested the ablation study and the GRNBoost2-based gene selection.

Appendix A Regularised estimation of sparse networks and covariances

Detailed review in the network-inference vignette.

- estimand: the precision $\mathbf{\Omega}_j$ of the within-type Gaussian; zeros are conditional independences, and partial correlations are $-\omega_{gg'}/\sqrt{\omega_{gg}\omega_{g'g'}}$
- three thresholds that are often conflated
  - the penalised estimator exists: with an off-diagonal penalty, the graphical lasso has a unique solution for any $\lambda > 0$ once every variance is positive, even when $G > n$ [126]
  - the unpenalised maximum-likelihood estimate needs $n > G$ [60]; on a fixed graph, the refit under its zeros needs n above the graph's maximum-likelihood threshold [127]
  - the graph is recovered when $n \asymp d^2 \log G$ for maximum degree d, under irrepresentability, bounded eigenvalues and a minimum signal; these conditions cannot be checked on real data, and they fail on models derived from gene expression [60, 61]
- estimators
  - partial correlations by inversion, $\hat{\mathbf{\Omega}} = S^{-1}$, with t tests and a positive-definite refit on the selected support; limited to $n > G$ [53]
  - graphical lasso [30, 128]: with a penalised diagonal the fitted variances become $S_{gg} + \lambda$, which biases the precision and the partial correlations without moving the empty-graph threshold $\max_{g \neq g'} |R_{gg'}|$; our pipeline fits the correlation matrix with `glasso(penalize.diagonal = FALSE)`
  - partial-correlation graphical lasso: penalises the partial correlations only, is scale invariant, and alternates between the partial correlations and the diagonal [129]
  - joint sparse regression (SPACE): node-wise regressions made symmetric, with the diagonal re-estimated from residual variances; better hub recovery [130]
  - D-trace loss: an alternative to the likelihood with a weaker irrepresentability condition [131]
  - de-sparsified graphical lasso with entrywise inference and global false discovery rate control [55, 57, 58], on nonparanormal scores [56]
  - Poisson–log-normal network: the precision of a latent Gaussian behind Poisson counts, free of the sampling noise that attenuates count correlations [28]
  - dense shrinkage: correlations shrunk towards zero with a closed-form intensity, variances kept; positive definite for any n and G [59]
- several cell types

  - shrunk covariances $\alpha S^{(j)} + (1-\alpha)\bar{S}$, shared support (group lasso) or sign-coherent edges (cooperative lasso) [132–134]
  - differential networks estimated directly, without either network [135]
  - fusion pulls rare types towards the dominant network, and removes the covariance differences that DeCovarT uses to separate types
- penalty selection: extended BIC [54], StARS [136], thresholded adaptive validation [137], or a graphical horseshoe instead of a path [138]

Appendix B A covariance-driven statistical simulator

B.1 Synthetic network simulation

Three families of random graphs generate the skeletons of the precision matrices.

- generalised random graph: given an arbitrary degree distribution, edges are placed so that the graph is no longer forced to be homogeneous [139]
- Preferential attachment: nodes are added one at a time and connect preferentially to well-connected nodes [26, 140]. The degree distribution follows a power law, which produces hubs; such scale-free graphs are often assumed for biological networks, although the evidence is debated [41, 141].
- Stochastic block model (affiliation model): a latent label assigns each node to a community, and edges are more likely within a block than between blocks [27].

Signed weights on the edges give a symmetric matrix $\boldsymbol{W}$, which is turned into a positive definite precision by the affine spectrum transformation

$$\boldsymbol{\Omega} = \boldsymbol{W} + \big(|\lambda_{\min}(\boldsymbol{W})| + u\big)\mathbf{I}_G, \qquad u > 0, \tag{B1}$$

which keeps the off-diagonal support and signs (Theorem 1) and sets the smallest eigenvalue of $\boldsymbol{\Omega}$ to u [28].

Theorem 1 (Invariance of the spectrum function by affine transformation) *Let X be any square matrix, and consider the affine transformation:*

$$Y = aX + bI$$

with a and b positive scalars, and I the identity matrix. Then Y has the same eigenvectors as X, and every eigenvalue λ of X becomes $a\lambda+b$: the spectrum is scaled by a and shifted by b.

Appendix C Review of deconvolution algorithms leveraging cell-type-specific variances

Detailed review in the review vignette.

C.1 Measurement versus expression models

- a measurement model $p(\boldsymbol{x} \mid \boldsymbol{\lambda})$ is separated from an expression model $p(\boldsymbol{\lambda})$; for UMI counts $x_{gc} \sim \text{Poisson}(s_c b_g m_{gc})$, and the capture b_g cancels within one protocol but not between a single-cell reference and a bulk library [62]
- overdispersion in UMI data is an expression property plus extrinsic noise, so a moment-based method needs no zero-inflated layer
- the size factor splits into a biological and a technical part; normalising by total counts removes the biological part, including cell size [63]
- early methods fitted unlogged microarray intensities of sorted populations [142–144], whose only second moment was a between-replicate variance

C.2 Errors in variables and moment estimators

- errors-in-variables reading: every method fits $\boldsymbol{y}_{\cdot i} = \hat{\boldsymbol{\mu}}\boldsymbol{p}_{\cdot i} + \boldsymbol{e}_i$ with latent profiles $\boldsymbol{x}_{\cdot j,i} = \boldsymbol{\mu}_{\cdot j} + \boldsymbol{u}_{\cdot j,i}$ and reference error $\hat{\boldsymbol{\mu}} = \boldsymbol{\mu} + \boldsymbol{v}$; *RNA-Sieve* and *MEAD* model $\boldsymbol{v}$ [10, 15], the others plug $\hat{\boldsymbol{\mu}}$ in
- the error law is parametric (DeCovarT, *RCTD*), asymptotic (*RNA-Sieve*, *MEAD*) or given by its first two moments only (*DECALS*, *MuSiC*, *DWLS*, *deMixSC*) [47]; the classical assumptions fail because the error variance $\sum_j p_{ji}^2 \boldsymbol{\Sigma}_j$ (DeCovarT) or the Poisson mean (*RCTD*) depends on $\boldsymbol{p}$
- our reading: $\boldsymbol{u}$ is a Berkson error weighted by p_{ji}; $\boldsymbol{v}$ is a classical error that DeCovarT ignores
- weighted NNLS (*MuSiC*, *DWLS*, *deMixSC*) and *Unico* are generalised-method-of-moments estimators: point estimates, and no interval without a variance model, a sandwich or a resampling unit [5, 6, 105]
- *deMixSC* (Guo et al. 2025): weighted NNLS with weight $w_{gi}^{-1} = \hat{y}_{gi}^2 + (y_{gi} - \hat{y}_{gi})^2 + 2$, combining the *DWLS* relative-error term and *MuSiC*'s residual variance; genes discordant between matched bulk and pseudo-bulk are down-weighted by $a^{-2} = 10^{-6}$; point estimates

C.3 Cell-type-specific moments given the composition

- with $\{\boldsymbol{p}_{\cdot i}\}$ known, the unknowns become the type moments or the sample-level profiles
- *CSNet*: gene-wise least squares for the means, regression of residual products on p_{ji}^2 for each covariance entry, then SCAD on the correlation; a marginal co-expression graph that needs many bulks of varying composition [89]
- *bMIND*: a $J \times J$ covariance across types within each gene, with an inverse-Wishart prior from single-cell summaries; proportions fixed beforehand [88]
- *EPIC-unmix* re-centres that prior on a first-step posterior, using the target samples twice [145]
- *Unico*: proportions given, $\text{Var}(y_{gi}) = \boldsymbol{p}_{\cdot i}^\top \boldsymbol{\Gamma}_g \boldsymbol{p}_{\cdot i} + \tau_g^2$ with a $J \times J$ covariance $\boldsymbol{\Gamma}_g$ across cell types, fitted by moments; sample-level profiles by a conditional expectation [105]

- a $J \times J$ covariance within a gene and DeCovarT's $G \times G$ covariance within a type answer different questions; no reviewed method fits both

C.4 Bayesian approaches

- *BLADE*: log-normal profiles with priors centred on single-cell moments, variational inference [17]
- *BEDwARS*: log-scale profiles around a noisy reference, Metropolis–Hastings, a cohort-level signature [18]
- *BayesPrism*: multinomial counts with a single-cell prior, Gibbs sampling, transcript fractions [19]
- *DSection*: Gaussian on linear intensities with a Dirichlet prior centred on a prior guess of the proportions [16]
- *gnSPADE*: reference-free topic model whose network prior acts on labels, not on expression [146]

C.5 Semi-reference and reference-free methods

- one unknown component: *DeMix* and *DeMixT* mix $\log_2$-normal components on the raw scale [147, 148]; *DeMixNB* uses an exact negative-binomial convolution [103]; *BayICE* mixes on the log scale, a mean regression rather than a convolution [102]; *ISOpureR* is a multinomial with a Dirichlet-shrunk tumour profile [85]
- reference-free with co-expression structure: *DSSC* fits similarity kernels built from the same bulk [149]; *GSNMF+* regularises the signature with a co-expression graph [92]; *swCAM* penalises sample-specific deviations with a nuclear norm, which implies a low-rank gene covariance [94]; all three use a Frobenius loss and ignore the mean–variance relationship of counts

C.6 Caveats shared across methods

- mixing is additive only on a linear scale; methods that mix logarithms fit a different model
- count and topic models, and any fit on library-normalised bulk, return transcript fractions; cell fractions need the cell sizes S_j
- independent cells give weight p_{ji}, a random population profile gives p_{ji}^2; a one-donor reference identifies only the first
- data reuse inflates apparent accuracy: empirical-Bayes re-centring on the target samples, similarity graphs built from the bulk being factorised, and priors from the benchmark's own single-cell source
- collinear reference means, perfect markers with gene-wise scales, or factorisations without separability leave proportions unidentified at any sample size

Appendix D Methodological details on DeCovarT

D.1 Derivation of the unconstrained likelihood

Definition 1 (Multivariate Gaussian distribution) A random vector $\boldsymbol{X} \in \mathbb{R}^G$ follows $\mathcal{N}_G(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ when its density is $\det(2\pi\boldsymbol{\Sigma})^{-1/2} \exp\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\}$, with mean $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{X}]$ and a positive definite covariance $\boldsymbol{\Sigma}$. The diagonal of $\boldsymbol{\Sigma}$ holds the variances of the genes, the off-diagonal entries their covariances, and $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$ is the precision matrix. If $\boldsymbol{\Sigma}$ is not positive definite, some linear combination of the genes has zero or negative variance, and the distribution is degenerate.

Remark 1 (Scaling and sums of independent Gaussian vectors) If $\boldsymbol{X} \sim \mathcal{N}_G(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and p is a constant, then $p\boldsymbol{X} \sim \mathcal{N}_G(p\boldsymbol{\mu}, p^2\boldsymbol{\Sigma})$. If $\boldsymbol{X}_1 \sim \mathcal{N}_G(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\boldsymbol{X}_2 \sim \mathcal{N}_G(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$ are independent, then $\boldsymbol{X}_1 + \boldsymbol{X}_2 \sim \mathcal{N}_G(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)$. Both follow from the characteristic function, and by induction they give the convolution (2) of J independent cell-type profiles.

Remark 2 (Equivariance of the DeCovarT maximum-likelihood estimate) DeCovarT is equivariant to any fixed, invertible, gene-wise affine transformation applied consistently to bulk expression, reference means and reference covariances. Let $\boldsymbol{D} = \mathrm{diag}(s_1, \dots, s_G)$ with $s_g \neq 0$ and $\boldsymbol{m} \in \mathbb{R}^G$. The map $\boldsymbol{y}^\star_{\cdot i} = \boldsymbol{D}^{-1}(\boldsymbol{y}_{\cdot i} - \boldsymbol{m})$, $\boldsymbol{\mu}^\star = \boldsymbol{D}^{-1}(\boldsymbol{\mu} - \boldsymbol{m}\mathbf{1}_J^\top)$, $\boldsymbol{\Sigma}^\star_j = \boldsymbol{D}^{-1}\boldsymbol{\Sigma}_j\boldsymbol{D}^{-1}$ preserves the linear convolution with the same $\boldsymbol{p}_{\cdot i}$, because $\mathbf{1}_J^\top \boldsymbol{p}_{\cdot i} = 1$. The Mahalanobis residual is unchanged and the log-determinant shifts by a constant, so the maximiser $\hat{\boldsymbol{p}}$ is unchanged; a gene-wise z-score computed once from $\boldsymbol{\mu}$ is a special case. Transformations estimated separately for each cell type or each bulk sample are not equivalent. A logarithm is not affine: by Jensen's inequality it changes the first and second moments, and $\log(\boldsymbol{\mu}\boldsymbol{p}) \neq (\log \boldsymbol{\mu})\boldsymbol{p}$, so the convolution is no longer linear; for that reason *CIBERSORT* also requires a non-log linear scale [2].

Remark 3 (Matrix calculus) For an invertible matrix $\boldsymbol{A} = \boldsymbol{A}(p)$ of a scalar p and constant matrices $\boldsymbol{U}$, $\boldsymbol{V}$: $\partial \det \boldsymbol{A}/\partial p = \det \boldsymbol{A}\ \mathrm{tr}(\boldsymbol{A}^{-1}\partial\boldsymbol{A}/\partial p)$, hence $\partial \log\det \boldsymbol{A}/\partial p = \mathrm{tr}(\boldsymbol{A}^{-1}\partial\boldsymbol{A}/\partial p)$; $\partial(\boldsymbol{U}\boldsymbol{A}\boldsymbol{V})/\partial p = \boldsymbol{U}(\partial\boldsymbol{A}/\partial p)\boldsymbol{V}$; $\partial\boldsymbol{A}^{-1}/\partial p = -\boldsymbol{A}^{-1}(\partial\boldsymbol{A}/\partial p)\boldsymbol{A}^{-1}$; and $\partial^2\boldsymbol{A}^{-1}/\partial p_j\partial p_\ell = \boldsymbol{A}^{-1}\{\boldsymbol{A}_j\boldsymbol{A}^{-1}\boldsymbol{A}_\ell - \boldsymbol{A}_{j\ell} + \boldsymbol{A}_\ell\boldsymbol{A}^{-1}\boldsymbol{A}_j\}\boldsymbol{A}^{-1}$, with $\boldsymbol{A}_j = \partial\boldsymbol{A}/\partial p_j$. For the convolution, $\partial\boldsymbol{\Sigma}/\partial p_j = 2p_j\boldsymbol{\Sigma}_j$ and $\partial\boldsymbol{r}/\partial p_j = -\boldsymbol{\mu}_{\cdot j}$, which give the score (4).

Differentiating (4) once more gives the Hessian $\mathbf{H} \in \mathbb{R}^{J\times J}$, with $\delta_{j\ell}$ the Kronecker delta:

$$\begin{aligned}\mathbf{H}_{j\ell} = &-\delta_{j\ell}\,\mathrm{tr}(\boldsymbol{\Omega}\boldsymbol{\Sigma}_j) + 2p_jp_\ell\,\mathrm{tr}(\boldsymbol{\Omega}\boldsymbol{\Sigma}_j\boldsymbol{\Omega}\boldsymbol{\Sigma}_\ell) - \boldsymbol{\mu}_{\cdot j}^\top\boldsymbol{\Omega}\boldsymbol{\mu}_{\cdot\ell} \\ &- 2p_\ell\,\boldsymbol{\mu}_{\cdot j}^\top\boldsymbol{\Omega}\boldsymbol{\Sigma}_\ell\boldsymbol{\Omega}\boldsymbol{r} - 2p_j\,\boldsymbol{\mu}_{\cdot\ell}^\top\boldsymbol{\Omega}\boldsymbol{\Sigma}_j\boldsymbol{\Omega}\boldsymbol{r} + \delta_{j\ell}\,\boldsymbol{r}^\top\boldsymbol{\Omega}\boldsymbol{\Sigma}_j\boldsymbol{\Omega}\boldsymbol{r} - 4p_jp_\ell\,\boldsymbol{r}^\top\boldsymbol{\Omega}\boldsymbol{\Sigma}_j\boldsymbol{\Omega}\boldsymbol{\Sigma}_\ell\boldsymbol{\Omega}\boldsymbol{r}.\end{aligned} \tag{D2}$$

Remark 4 (Expected Fisher information and Wald intervals) Taking expectations of (D2) under the generative model, with $\mathbb{E}[\boldsymbol{r}] = \mathbf{0}$ and $\mathbb{E}[\boldsymbol{r}\boldsymbol{r}^\top] = \boldsymbol{\Sigma}(\boldsymbol{p})$, cancels the determinant traces against the residual traces and returns $\mathbb{E}[-\mathbf{H}] = I(\boldsymbol{p})$, with

$$I(\boldsymbol{p})_{j\ell} = \boldsymbol{\mu}_{\cdot j}^\top\boldsymbol{\Omega}(\boldsymbol{p})\boldsymbol{\mu}_{\cdot\ell} + 2p_jp_\ell\,\mathrm{tr}\big(\boldsymbol{\Omega}(\boldsymbol{p})\boldsymbol{\Sigma}_j\boldsymbol{\Omega}(\boldsymbol{p})\boldsymbol{\Sigma}_\ell\big). \tag{D3}$$

The identity holds only with the factor $\frac{1}{2}$ on both terms of (3); doubling the log-determinant, as an early version of the implementation did, breaks it. In log-ratio coordinates $\boldsymbol{\rho}$, the Cramér–Rao bound is $\mathrm{Var}(\hat{\boldsymbol{\rho}}) \succeq I_{\boldsymbol{\rho}}^{-1}$ with $I_{\boldsymbol{\rho}} = \mathbf{J}_\psi^\top I(\boldsymbol{p}) \mathbf{J}_\psi$, and the delta method maps it back to the simplex:

$$\mathrm{Var}(\hat{\boldsymbol{p}}) = \mathbf{J}_\psi I_{\boldsymbol{\rho}}^{-1} \mathbf{J}_\psi^\top. \tag{D4}$$

This is the matrix returned by `vcov()` on a fitted DeCovarT object. It is not an ordinary least-squares residual variance: goodness of fit is the convolution log-likelihood, and the estimator does not forecast bulk expression.

D.2 Derivation of the constrained likelihood on log-ratio charts

Full derivations, worked examples and a figure in the generative-model vignette.

D.2.1 Four charts of the simplex

- constrained optimisation can eliminate the constraints by reparametrisation, enforce them softly with penalties or barriers, or enforce feasibility exactly by projections, Lagrangian or Karush–Kuhn–Tucker (KKT) conditions methods; reparametrisation and penalties are simple but can be poorly conditioned, barrier and interior-point methods suit convex problems, and Lagrangian methods (augmented Lagrangian, sequential quadratic programming) are the most general
- DeCovarT reparametrises with a chart of the open simplex; the derivatives of Appendix D.1 are ambient likelihood and Hessian and are mapped to the chart by the chain rule
- softmax of J scores $\boldsymbol{z}$: $\boldsymbol{p} = \mathrm{softmax}(\boldsymbol{z}) = \exp(\boldsymbol{z}) / \mathbf{1}^\top \exp(\boldsymbol{z})$; it is over-parametrised, $\mathrm{softmax}(\boldsymbol{z} + t\mathbf{1}) = \mathrm{softmax}(\boldsymbol{z})$, and its Jacobian $\boldsymbol{S}(\boldsymbol{p}) = \mathrm{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\top$ satisfies $\boldsymbol{S}(\boldsymbol{p})\mathbf{1} = \mathbf{0}$, so a Newton system in $\boldsymbol{z}$ is singular
- the three log-ratio charts remove that direction [8, 35]: the additive log-ratio $\boldsymbol{\rho}^{\mathrm{alr}} = \boldsymbol{D}_r \log \boldsymbol{p}$, whose rows $\boldsymbol{e}_j^\top - \boldsymbol{e}_r^\top$ pin a reference type r; the centred log-ratio $\boldsymbol{\xi} = \boldsymbol{H}_J \log \boldsymbol{p}$, with $\boldsymbol{H}_J = \mathbf{I}_J - J^{-1}\mathbf{1}\mathbf{1}^\top$, whose J coordinates sum to zero; and the isometric log-ratio

$$p_j = \psi_j(\boldsymbol{\rho}) = \frac{e^{z_j}}{\sum_{\ell=1}^{J} e^{z_\ell}}, \qquad \boldsymbol{z} = \mathbf{V}\boldsymbol{\rho}, \qquad \boldsymbol{\rho} = \mathbf{V}^\top \log \boldsymbol{p}, \tag{D5}$$

  where $\mathbf{V}$ is a $J \times (J-1)$ Helmert basis; the additive log-ratio is the softmax of $(\boldsymbol{\rho}^{\mathrm{alr}}, 0)$ when $r = J$
- $\boldsymbol{\psi}$ is a C^2-diffeomorphism between $\mathbb{R}^{J-1}$ and the open simplex; the log-sum-exp trick evaluates it and its derivatives without overflow

D.2.2 The Helmert basis

- column $j = 1, \ldots, J-1$ of $\mathbf{V}$ is $\boldsymbol{v}_j = (1, \ldots, 1, -j, 0, \ldots, 0)^\top / \sqrt{j(j+1)}$, with j ones; then $\mathbf{V}^\top\mathbf{V} = \mathbf{I}_{J-1}$, $\mathbf{V}^\top\mathbf{1} = \mathbf{0}$ and $\mathbf{V}\mathbf{V}^\top = \boldsymbol{H}_J$ [21, 35]

Table D1 Four charts of the open simplex. κ is the condition number; $\boldsymbol{A}_r = \boldsymbol{D}_r\mathbf{V}$ maps ILR to ALR coordinates.

Property	Softmax	ALR	CLR	ILR
Free coordinates	J, one redundant	$J-1$	J summing to zero	$J-1$
Privileged cell type	none	reference r	none	none
Isometry for the Aitchison distance	no	no (oblique basis)	yes	yes
Change of convention	–	oblique, $\kappa(\boldsymbol{A}_r) = \sqrt{J}$	–	orthogonal rotation
Hessian conditioning	singular	depends on r	singular	invariant

- each coordinate is a balance, $\rho_j = \sqrt{j/(j+1)}\,\log\{g(p_1,\dots,p_j)/p_{j+1}\}$ with g the geometric mean: the first j types against type $j+1$
- construction: Gram–Schmidt applied to the successive differences $\boldsymbol{e}_j - \boldsymbol{e}_{j+1}$, which span $\mathbf{1}^\perp$, returns exactly these columns; every other orthonormal basis of $\mathbf{1}^\perp$ is $\mathbf{V}\boldsymbol{Q}$ with $\boldsymbol{Q}^\top\boldsymbol{Q} = \mathbf{I}_{J-1}$, so the order of the cell types fixes which balances are reported, not the estimate

D.2.3 Derivatives, invariance and conditioning

- Jacobian and second derivatives:

$$\mathbf{J}_\psi = \boldsymbol{S}(\boldsymbol{p})\mathbf{V}, \qquad \frac{\partial^2 p_j}{\partial z_\ell \partial z_{\ell'}} = p_j\big[(\delta_{j\ell} - p_\ell)(\delta_{j\ell'} - p_{\ell'}) - p_\ell(\delta_{\ell\ell'} - p_{\ell'})\big], \tag{D6}$$

  and $\partial^2 p_j/\partial\rho_a\partial\rho_b = \sum_{\ell,\ell'} V_{\ell a}V_{\ell' b}\,\partial^2 p_j/\partial z_\ell \partial z_{\ell'}$, a third-order tensor symmetric in (a,b) by Schwarz's theorem
- chain rules, in $\mathbb{R}^{J-1}$ and $\mathbb{R}^{(J-1)\times(J-1)}$:

$$\nabla_{\boldsymbol{\rho}}\ell = \mathbf{J}_\psi^\top \nabla_{\boldsymbol{p}}\ell, \qquad \nabla^2_{\boldsymbol{\rho}}\ell = \mathbf{J}_\psi^\top \mathbf{H}\mathbf{J}_\psi + \sum_{j=1}^{J} \frac{\partial \ell}{\partial p_j}\nabla^2_{\boldsymbol{\rho}} p_j \tag{D7}$$

  with $\mathbf{H}$ from (D2); the second term vanishes at an interior KKT point (below) but not in general
- statistical invariance: for any invertible linear change of coordinates $\boldsymbol{\rho}' = \boldsymbol{A}\boldsymbol{\rho}$, the maximiser, the maximised log-likelihood and the delta-method covariance are unchanged, since $(\mathbf{J}_\psi\boldsymbol{A}^{-1})(\boldsymbol{A}^{-\top} I_{\boldsymbol{\rho}} \boldsymbol{A}^{-1})^{-1}(\mathbf{J}_\psi\boldsymbol{A}^{-1})^\top = \mathbf{J}_\psi I_{\boldsymbol{\rho}}^{-1}\mathbf{J}_\psi^\top$; the pure Newton step is also affine invariant [36]
- numerical invariance: a change of ILR basis gives $\mathbf{H}_{\boldsymbol{\rho}^\star} = \boldsymbol{Q}^\top\mathbf{H}_{\boldsymbol{\rho}}\boldsymbol{Q}$, with the same eigenvalues; the ALR map $\boldsymbol{A}_r = \boldsymbol{D}_r\mathbf{V}$ has $J-2$ singular values equal to 1 and

one equal to $\sqrt{J}$, along the log-abundance of the reference $\mathbf{V}^\top \boldsymbol{e}_r$, so $\kappa(\mathbf{H}_{\boldsymbol{\rho}})/J \le \kappa(\boldsymbol{A}_r^{-\top}\mathbf{H}_{\boldsymbol{\rho}}\boldsymbol{A}_r^{-1}) \le J\,\kappa(\mathbf{H}_{\boldsymbol{\rho}})$

- consequence: gradient norms, damping, stopping rules and the floating-point accuracy of $I_{\boldsymbol{\rho}}^{-1}$ are the same in every ILR basis but change with the ALR reference, most when the reference type is rare or poorly identified

D.2.4 Lagrangian and Karush–Kuhn–Tucker conditions

- without a chart, the problem is $\max_{\boldsymbol{p}} \ell(\boldsymbol{p})$ subject to $\mathbf{1}^\top \boldsymbol{p} = 1$ and $\boldsymbol{p} \ge \mathbf{0}$, with Lagrangian $\mathcal{L}(\boldsymbol{p}, \lambda, \boldsymbol{\nu}) = \ell(\boldsymbol{p}) - \lambda(\mathbf{1}^\top \boldsymbol{p} - 1) + \boldsymbol{\nu}^\top \boldsymbol{p}$
- the constraints are linear, so a constraint qualification holds and every local maximiser satisfies the Karush–Kuhn–Tucker (KKT) conditions conditions [36]:

$$\nabla_{\boldsymbol{p}} \ell(\hat{\boldsymbol{p}}) - \lambda \mathbf{1} + \boldsymbol{\nu} = \mathbf{0}, \qquad \mathbf{1}^\top \hat{\boldsymbol{p}} = 1, \qquad \boldsymbol{\nu} \ge \mathbf{0}, \qquad \nu_j \hat{p}_j = 0 \tag{D8}$$

- complementary slackness, $\nu_j \hat{p}_j = 0$, says which constraints determine the optimum: either $\hat{p}_j > 0$, the constraint is inactive and $\nu_j = 0$, or $\hat{p}_j = 0$, the constraint binds and ν_j may be positive

> The reading of complementary slackness follows teaching slides (*Optimization in Statistics*, StatHarbor Analytics, 2026; local extraction under `.firecrawl/extract/`), which are not in Zotero; add them, or keep Boyd and Vandenberghe only.

- interior point: $\boldsymbol{\nu} = \mathbf{0}$ and $\partial \ell / \partial p_j = \lambda$ for every j, with $\lambda = \hat{\boldsymbol{p}}^\top \nabla_{\boldsymbol{p}} \ell(\hat{\boldsymbol{p}})$; boundary point: $\partial \ell / \partial p_j \le \lambda$ for every absent type
- the chart removes the multiplier: since $\boldsymbol{S}(\boldsymbol{p})\mathbf{1} = \mathbf{0}$, $\nabla_{\boldsymbol{\rho}} \ell = \mathbf{V}^\top \boldsymbol{S}(\boldsymbol{p}) \{\nabla_{\boldsymbol{p}} \ell - \lambda(\boldsymbol{p})\mathbf{1}\}$ with $\lambda(\boldsymbol{p}) = \boldsymbol{p}^\top \nabla_{\boldsymbol{p}} \ell$, and $\|\nabla_{\boldsymbol{\rho}} \ell\| = \|\boldsymbol{S}(\boldsymbol{p}) \nabla_{\boldsymbol{p}} \ell\|$; hence $\nabla_{\boldsymbol{\rho}} \ell = \mathbf{0}$ if and only if $\nabla_{\boldsymbol{p}} \ell = \lambda \mathbf{1}$: stationary points in $\boldsymbol{\rho}$ are exactly the interior KKT points, and the score norm does not depend on the basis
- reduced Hessian: at an interior KKT point the second term of (D7) vanishes, because $\sum_j \nabla^2_{\boldsymbol{\rho}} p_j = \mathbf{0}$, and $\nabla^2_{\boldsymbol{\rho}} \ell = \mathbf{J}_\psi^\top \nabla^2_{\boldsymbol{pp}} \mathcal{L}\, \mathbf{J}_\psi$; the columns of $\mathbf{J}_\psi$ span the tangent space $\{\boldsymbol{t} : \mathbf{1}^\top \boldsymbol{t} = 0\}$, so a negative-definite ILR Hessian is the second-order sufficient condition of the constrained problem
- limit: for finite $\boldsymbol{\rho}$ every $p_j > 0$, so the inequality multipliers never activate; an estimate on a face lies at infinity in $\boldsymbol{\rho}$ and is handled by restricted fits and boundary tests (Appendix D.5)

D.3 Alternative alignment strategies

Derivations in the single-cell alignment vignette.

- reference regimes
  - sorted profiles: one random population vector per type scaled by p_{ji}, so Cov $= \sum_j p_{ji}^2 \boldsymbol{\Sigma}_{B,j}$, the released DeCovarT covariance
  - one single-cell replicate: $n_i^{-1} \sum_j p_{ji} \boldsymbol{\Sigma}_{W,j} + \boldsymbol{\Sigma}_{\mathrm{obs},i}$ for the bulk per cell

  - $R \geq 2$ replicates: method-of-moments $\hat{\boldsymbol{\Sigma}}_{B,j} = \boldsymbol{S}_{B,j} - \hat{\boldsymbol{\Sigma}}_{W,j}/\bar{C}_j$, projected on the positive semidefinite cone, and the three-layer law $\sum_j p_{ji}^2 \boldsymbol{\Sigma}_{B,j} + n_i^{-1} \sum_j p_{ji} \boldsymbol{\Sigma}_{W,j} + \boldsymbol{\Sigma}_{\text{obs},i}$; small cohorts support a scalar or sparse $\boldsymbol{\Sigma}_{B,j}$, not a dense one
  - one identity covers all three: $\text{Cov}(\sum_{r=1}^{a} \boldsymbol{X}_r) = a[1 + (a-1)\varrho]\boldsymbol{\Sigma}$
- strategy 1, cell-count convolution: targets $\boldsymbol{m}_{\cdot i} \in \mathbb{N}^J$ with covariance linear in $\boldsymbol{m}_{\cdot i}$, as in *MuSiC* before library division [5]; the variance identifies the number of cells, as in *RNA-Sieve* [10]
- strategy 2, averaged mixture on the simplex: $\mathcal{N}_G(a_i \tilde{\boldsymbol{\mu}} \boldsymbol{p}_{\cdot i}, a_i^2[\boldsymbol{S}_B(\boldsymbol{p}_{\cdot i}) + n_i^{-1} \boldsymbol{S}_W(\boldsymbol{p}_{\cdot i})] + \boldsymbol{D}_i)$; released DeCovarT is the limit $n_i \to \infty$, $a_i = 1$; *DECALS* keeps $\boldsymbol{S}_B$ only [14]; strategies 1 and 2 are one Gaussian family
- strategy 3, library-size closure: the closed bulk has a singular covariance and independent genes become negatively correlated; the target becomes RNA fractions [8]
- strategy 4, plug-in scales: technical size factor, S_j, platform ratio $\tilde{\boldsymbol{b}}$ or library offset estimated from auxiliary data, as in *RCTD* [12]; the variance of $\hat{\boldsymbol{p}}$ gains $I_{pp}^{-1} I_{p\eta} \boldsymbol{V}_\eta I_{\eta p} I_{pp}^{-1}$ for a plug-in $\hat{\boldsymbol{\eta}}$ of covariance $\boldsymbol{V}_\eta$
- why the model of Section 3.7.4 was chosen: strategy 1 needs a lattice search over integer cell counts for rare types; the full strategy 2 needs $R \geq 2$ donors; strategy 3 breaks the additive mixture; a gene-wise $\tilde{\boldsymbol{b}}$ needs a paired cohort; with one donor, strategies 1 and 2 meet in the linear layer, with the two scales $c_i = a_i n_i$ and $\phi_i = a_i^2 n_i$ estimated jointly and S_j kept in $\tilde{\boldsymbol{\mu}}$

D.4 Numerical optimisation algorithms

- mean-only baselines: NNLS and quadratic programming (`lsei`) on $\|\boldsymbol{y} - \boldsymbol{\mu}\boldsymbol{p}\|^2$ [4, 9]
- likelihood solvers: L-BFGS-B with box constraints and renormalisation [112]; projected gradient; Newton–Raphson (default, `stats::nlminb`) and Marquardt–Levenberg on the ILR chart with the analytic Hessian [32–34]; simulated annealing as a global reference
- the Marquardt–Levenberg algorithm [32] moves between steepest ascent (first order) and Newton–Raphson (second order) by inflating the diagonal of the Hessian: it takes Newton-like steps where the quadratic model is reliable and short gradient steps where it is not; the `marqLevAlg` implementation adds the relative distance to the maximum, which separates maxima from saddle points [33]
- that stopping rule suits a balanced ridge and fires early on a rare-type tail
- failure is split into numerical failure (no finite estimate) and theoretical failure (estimate below the likelihood at the truth); stationarity is measured by the projected-score residual, valid on faces

Report run time and memory per solver from the fig02 and fig03 books.

D.5 Maximum-likelihood properties and confidence intervals

Proofs and simulations in the maximum-likelihood vignette.

D.5.1 Existence, uniqueness and identifiability

- existence: with symmetric positive-definite $\boldsymbol{\Sigma}_j$, $\boldsymbol{\Sigma}(\boldsymbol{p})$ is positive definite on the closed simplex, so the log-likelihood is continuous on a compact set and attains its supremum, possibly on a face
- non-uniqueness: the model is heteroscedastic; with $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \mathbf{I}_3$, $\boldsymbol{\mu}_{\cdot 1} = (2, 1, 1)^\top$, $\boldsymbol{\mu}_{\cdot 2} = (1, 2, 6/5)^\top$ and $\boldsymbol{y} = \boldsymbol{\mu}(1/2, 1/2)^\top$, the Hessian (D2) at $\boldsymbol{p} = (1/5, 3/10)$ has eigenvalues of about -654.5 and $+53.9$; the log-likelihood is therefore not concave, in $\boldsymbol{p}$ or in the ILR chart, and a converged solver certifies a local maximum only
- a single bulk profile provides no replication of its own composition, so every asymptotic statement below is taken in the number of replicate samples sharing one composition
- population uniqueness: the expected log-likelihood is maximised only at the true composition under identifiability, because the gap is a Kullback–Leibler divergence [150]
- ILR versus *MEAD*: *MEAD* truncates or softplus-transforms its estimate, which is inconsistent for a finite softplus constant [15]; the softmax chart gives exact positivity with no tuning, and a true zero sits at infinity in ILR coordinates

D.5.2 Wald intervals

- $I_{\boldsymbol{\rho}} = \mathbf{J}_\psi^\top I(\boldsymbol{p}) \mathbf{J}_\psi$ with $\mathbf{J}_\psi = (\operatorname{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\top)\mathbf{V}$, and $\hat{\boldsymbol{\rho}} \overset{a}{\sim} \mathcal{N}(\boldsymbol{\rho}_0, I_{\boldsymbol{\rho}}^{-1})$
- delta method: $\operatorname{Var}(\hat{\boldsymbol{p}}) \approx \mathbf{J}_\psi I_{\boldsymbol{\rho}}^{-1} \mathbf{J}_\psi^\top$; symmetric, not invariant to reparametrisation [37], and outside $[0, 1]$ for rare types in unbalanced mixtures; truncation hides the problem
- back-transformed routes: a logit interval with $\operatorname{SE}(\operatorname{logit} \hat{p}_j) = \operatorname{SE}(\hat{p}_j)/\{\hat{p}_j(1 - \hat{p}_j)\}$, or quantiles of $\psi(\boldsymbol{\rho}^{(b)})$ with $\boldsymbol{\rho}^{(b)} \sim \mathcal{N}(\hat{\boldsymbol{\rho}}, I_{\boldsymbol{\rho}}^{-1})$; both respect the bounds, neither guarantees coverage, and both are undefined on a face
- near a face the Wald statistic can decrease as the estimate moves away from the null (Hauck–Donner effect), detected by ratios of Wald, score and likelihood-ratio statistics [39]

D.5.3 Finite-sample bias-corrected intervals

- Firth's penalty $\ell + \frac{1}{2} \log\det I$ removes the first-order bias of canonical exponential families [151]; in the ILR chart $\det I$ vanishes at the faces, so the penalised estimate is finite and its Wald interval defined; for this curved Gaussian family the bias removal needs a simulation check
- the plug-in precision is biased, $\mathbb{E}[\hat{\boldsymbol{\Sigma}}_j^{-1}] = \frac{\nu_j}{\nu_j - G - 1} \boldsymbol{\Sigma}_j^{-1}$ for $\nu_j = C_j - 1 > G + 1$; the factor $(\nu_j - G - 1)/\nu_j$ corrects it, and *DECALS* corrects its own covariance estimator [14]

- sandwich covariance $\boldsymbol{H}^{-1}\boldsymbol{K}\boldsymbol{H}^{-1}$ under a misspecified covariance, as in *RNA-Sieve* [10]; with one bulk column, the log-likelihood factorises over whitened genes, $\ell = \sum_g \ell_g$ with $\boldsymbol{e} = \boldsymbol{L}^{-1}(\boldsymbol{y} - \boldsymbol{\mu}\boldsymbol{p})$, so $\hat{\boldsymbol{K}} = \sum_g \nabla\ell_g \nabla\ell_g^\top$ and the robust Firth variance is $\boldsymbol{H}_{\mathrm{F}}^{-1}\hat{\boldsymbol{K}}\boldsymbol{H}_{\mathrm{F}}^{-1}$
- this sandwich differs from the variational one used for Poisson–log-normal models [152]

Implement the gene-factorised sandwich; check its dependence on gene order under misspecification.

D.5.4 Profile likelihood, boundary tests and bootstraps

- profile likelihood: $D_j(c) = 2\{\ell(\hat{\boldsymbol{p}}) - \max_{p_j = c} \ell(\boldsymbol{p})\} \rightsquigarrow \chi^2_1$ [38], invariant to the chart, asymmetric and inside the simplex
- boundary nulls: the limit is the projection of a Gaussian score on the tangent cone of the simplex [22]; with q active and s interior constraints and orthogonal information blocks, $D \rightsquigarrow \sum_{i=0}^{q} \binom{q}{i} 2^{-q} \chi^2_{s+i}$, and $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ for one absent type [153]
- without orthogonality the weights are calibrated by a restricted parametric bootstrap; the null $p_j = 0$ cannot be written in ILR coordinates, so restricted fits substitute the constrained value directly
- parametric bootstrap from $\mathcal{N}_G(\boldsymbol{\mu}\hat{\boldsymbol{p}}, \boldsymbol{\Sigma}(\hat{\boldsymbol{p}}))$; reference bootstrap over donors or cells, which rebuilds $(\boldsymbol{\mu}, \boldsymbol{\Sigma}_j)$ and refits [154]
- generalised-linear-model view: DeCovarT models the mean through an inverse link on the parameter instead of transforming the response [47], unlike transformation-based methods such as *DeOPUS* [76]; its quasi-score yields the planned intervals of Section 3.3

Table D2 Which interval or test to report for DeCovarT proportions.

Procedure	Calibration	Simplex faces	Status
Wald, delta method	$\mathbf{J}_\psi I_{\boldsymbol{\rho}}^{-1} \mathbf{J}_\psi^\top$; may leave $[0, 1]$	unusable	implemented
Logit or chart Wald	back-transformed bounds	undefined	partly
Firth-penalised Wald	Jeffreys-penalised likelihood	defined	planned
Sandwich	gene-factorised scores	interior only	planned
Profile likelihood	χ^2_1 on the deviance	inside simplex	implemented
Boundary LRT	$\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$	designed for it	implemented
Parametric bootstrap	refit on simulated bulks	automatic	implemented
Reference bootstrap	resample donors or cells	automatic	implemented

Appendix E Special cases of the generative model

DeCovarT's residual covariance $\boldsymbol{\Sigma}(\boldsymbol{p}_{\cdot i}) = \sum_j p_{ji}^2 \boldsymbol{\Sigma}_j$ depends on the unknown composition. Freezing it at the barycentre $p_j = 1/J$ gives a single residual covariance,

$$\mathrm{Cov}(\boldsymbol{y}_{\cdot i} \mid \boldsymbol{\mu}) = \frac{1}{J^2} \sum_{j=1}^{J} \boldsymbol{\Sigma}_j, \tag{E9}$$

so the variance of the purified profiles no longer depends on their mixing weights. The fit is then generalised least squares on the linear mean (1). Weighted least squares is the diagonal case $\boldsymbol{\Omega} = \mathrm{diag}(w_1, \ldots, w_G)$, for which *DWLS* supplies dampened gene weights [6]. Interaction monomials $p_j p_\ell$ in the mean (a Scheffé expansion) can be added on the same residual, at the cost of many extra parameters.

Theorem 2 (Generalised least squares) *Consider $\boldsymbol{y}_{\cdot i} = \boldsymbol{\mu}\boldsymbol{p}_{\cdot i} + \boldsymbol{\varepsilon}$ with $\mathbb{E}(\boldsymbol{\varepsilon} \mid \boldsymbol{\mu}) = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\varepsilon} \mid \boldsymbol{\mu}) = \boldsymbol{\Sigma}$, a known non-singular covariance. The estimate that minimises the squared Mahalanobis norm of the residuals is $\hat{\boldsymbol{p}}^{\mathrm{GLS}} = (\boldsymbol{\mu}^\top \boldsymbol{\Omega} \boldsymbol{\mu})^{-1} \boldsymbol{\mu}^\top \boldsymbol{\Omega} \boldsymbol{y}_{\cdot i}$, with $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$. It is the best linear unbiased estimator of $\boldsymbol{p}_{\cdot i}$, and it equals the unconstrained maximum-likelihood estimate under a Gaussian error. A Cholesky factor $\boldsymbol{\Sigma} = \boldsymbol{C}\boldsymbol{C}^\top$ reduces the problem to ordinary least squares on $\boldsymbol{C}^{-1}\boldsymbol{y}_{\cdot i}$, which removes heteroscedasticity and residual correlation. The simplex constraint is imposed afterwards, or through the chart* (D5).

Appendix F Performance metrics

Formulas and simulations in the synthetic-scenarios vignette. Notation: $\boldsymbol{e} = \hat{\boldsymbol{p}} - \boldsymbol{p}^\star$.

- framing: ADEMP, with absolute, angular, association, inference and optimisation scores kept in separate tables and no composite score [25, 155]
- regression-type scores
  - on the simplex $\|\boldsymbol{e}\|_1 \leq 2$; total variation $\frac{1}{2}\|\boldsymbol{e}\|_1 \in [0, 1]$ is the default absolute error, and the mean absolute error $2d_{\mathrm{TV}}/J$ adds nothing at fixed J
  - RMSE $(J^{-1}\sum_j e_j^2)^{1/2}$ and the maximum error $\|\boldsymbol{e}\|_\infty$
  - pitfall: a miss of 0.05 counts the same at $p_j = 0.5$ and at $p_j = 0.01$
  - cross-sample Pearson correlation measures ranking, not calibration, and is undefined when the true composition is constant within a scenario
- compositional scores
  - Aitchison distance $\|\mathrm{clr}\,\boldsymbol{p}^\star - \mathrm{clr}\,\hat{\boldsymbol{p}}\|_2$, equal to the Euclidean distance between ILR coordinates [8, 21]
  - pitfall: undefined at zero; with a machine-epsilon floor, $\boldsymbol{p}^\star = (0.98, 0.01, 0.01)$ and $\hat{\boldsymbol{p}} = (1, 0, 0)$ give a total variation of 0.02 but an Aitchison distance near 26, so averages count exact zeros rather than accuracy
  - Kullback–Leibler divergence is infinite as soon as $\hat{p}_j = 0 < p_j^\star$; the Jensen–Shannon divergence stays finite with zeros
- Monte Carlo summaries

  - bias, empirical standard deviation, Monte Carlo RMSE and the ratio of model standard error to empirical standard deviation
  - coverage has target $1-\alpha$, with Monte Carlo standard error $\sqrt{\hat{\pi}(1-\hat{\pi})/B}$ (0.022 at $B = 100$) and a Wilson interval [65]; report the mean width, because wide intervals inflate coverage
- failure categories: theoretical success; theoretical failure with numerical success; numerical failure

Table F3 Accuracy metrics for $\hat{\boldsymbol{p}}$ on the simplex, with $\boldsymbol{e} = \hat{\boldsymbol{p}} - \boldsymbol{p}^\star$.

Metric	Range	Zeros and rare types	Role
Total variation $\frac{1}{2}\|\boldsymbol{e}\|_1$	$[0, 1]$	finite; absolute scale	primary
RMSE	$[0, \sqrt{2/J}]$	finite; abundant types dominate	primary
Maximum error $\|\boldsymbol{e}\|_\infty$	$[0, 1]$	finite	primary
Pearson r_j across samples	$[-1, 1]$	undefined if truth constant	per type
Aitchison distance	$[0, \infty)$	undefined at zero	secondary
Jensen–Shannon	$[0, \log 2]$	finite with zeros	not implemented
Coverage	$[0, 1]$	Wald invalid near a face	primary

Appendix G Supplementary main simulation plots

G.1 Bivariate plot books

Plot books of the $G = 2$, $J = 2$ design ($N = 500$ bulks, 972 scenarios) are in the fig02 output directory. Density pages, under `density_visualisations/`, use the correlation corners $(\rho_1, \rho_2) \in \{(0, 0), (-0.8, -0.8), (0.8, 0.8), (-0.8, 0.8)\}$ and twelve meta-scenarios (centroid gap × variance × composition).

- `purified_density.pdf`: the two cell-type Gaussians before mixing
- `bulk_density.pdf`: the convolution that DeCovarT fits
- `loglik_surface_p.pdf`: log-likelihood of one bulk on the simplex; colour is $\log_{10}(L/\max L)$
- `loglik_ilr_profile.pdf`: the same surface as a curve in the single ILR coordinate
- `loglik_expected_surface_p.pdf`, `loglik_expected_ilr_profile.pdf`: expected log-likelihood at the true $\boldsymbol{p}$, not one draw
- `loglik_rgl.pdf`: the same relative likelihood as a 3D surface; white marker at $\boldsymbol{p}^\star$, red marker at the numerical maximum

Performance pages, under `performance_visualisations/`:

- `heatmap_rmse.pdf`, `heatmap_mae.pdf`, `heatmap_aitchison.pdf`: tiles on (ρ_1, ρ_2); one page per meta-scenario; seven solver panels
- `forest.pdf`: ADEMP dot-whisker for the convolution solvers [25, 156]

  - solid whisker: 1.96 times the expected-Fisher Wald standard error at the true composition
  - dashed whisker: empirical Monte Carlo standard deviation
  - labels: RMSE and coverage of that cell type
- `raincloud.pdf`: full Monte Carlo law of $\hat{p}$, including pile-up at 0 and 1
- `dotsinterval.pdf`: the same quantiles without a kernel density on $[0, 1]$
- `qq_normal.pdf`: whitened ILR coordinate against $N(0, 1)$
- `qq_chi2.pdf`: squared Mahalanobis distance against χ^2_1
- `ks_normal.pdf`: Kolmogorov–Smirnov statistic versus $N(0, 1)$
- `similarity.pdf`: Pearson correlation of paired $\hat{\boldsymbol{p}}$ inside one correlation corner; Ward clustering of $1 - r$
- `solver_dots.pdf`: the same 9×9 grid; colour is mean RMSE, size is mean Aitchison distance

Map one metric per plot: the Aitchison size channel is dominated by exact zeros.

- `kkt_raincloud.pdf`: projected-score residual
- `convergence_stacked.pdf`: theoretical success, numerical trouble, theoretical failure
- `runtime.pdf`, `memory.pdf`: $\log_{10}$ seconds and $\log_{10}$ MiB
- heatmaps: where on (ρ_1, ρ_2) a mean-only fit breaks, at fixed centroid gap
- large gap and $\rho_j = 0$: the mean map is injective, and every solver is accurate
- small gap: the least-squares residual is a ridge; NNLS and `lsei` snap to vertices
- rainclouds: bias, skew and boundary mass that a mean-plus-interval hides
- forests: whether the Wald standard error matches the Monte Carlo spread
- likelihood curves: a flat ILR profile means a rare type or colliding means
- Q–Q and Kolmogorov–Smirnov: the $N = 1$ normal approximation in ILR coordinates; Wilks does not apply on a face [22, 38]

G.2 Covariance-driven plot books

Plot books of the $G = 20$, $J = 3$ design (36 scenarios) are in the fig03 output directory. Density pages, under `density_visualisations/`:

- `mean_signature.pdf`: heatmap of $\boldsymbol{\mu}$ before scaling
- `network_topologies.pdf`: graph skeletons, one page per overlap target
- `scenario_metrics.pdf`: realised overlap, entropy, conditioning and related descriptors
- `latent_projections.pdf`: two-dimensional views of the convolution (Thomson, MCFA, `MclustDR`)
- `pairwise_hellinger.pdf`: pairwise discrepancy of the purified laws
- `loglik_surface_p.pdf`, `loglik_expected_surface_p.pdf`: one-draw and expected log-likelihood on the additive log-ratio plane
- `loglik_rgl.pdf`: 3D relative likelihood; same markers as in the bivariate book

Performance pages, under `performance_visualisations/`, follow Appendix G.1, with five solvers and no MAE tile book.

- signature heatmap: types 1 and 2 are the collinear pair; type 3 is the separated column
- networks: scale-free versus stochastic block model on types 1 and 2
- descriptor strip: overlap target and $H^\star$ against the design
- forests at high overlap: whether an ILR solver splits types 1 and 2 when `lsei` and *CIBERSORT* have only the mean
- rainclouds at $H^\star = 0.1$: boundary pile-up of the rare weight
- latent projections: whether a linear plane still shows three clouds
- likelihood surfaces: ridges in the two-dimensional ILR chart

Appendix H Supplementary simulations

The designs below are specified in the supplementary vignettes; none reports results yet.

H.1 Sensitivity to initial estimates

- the one-sample log-likelihood is not concave; a start near a face can stall second-order solvers on a plateau
- toy design: $G = 3$, $J = 2$, $\boldsymbol{p} = (0.6, 0.4)$, with Marquardt–Levenberg and Newton–Raphson started from three Dirichlet$(1, 1)$ draws; a spread of several percentage points between starts flags a trapped solver
- factorial design: $J = 2$, $G = 2$, low cosine distance between the mean columns, $\rho \in \{-0.8, 0, 0.8\}$, 1000 replicates; expected: the fraction of starts that reach the global mode falls as the mean columns approach each other
- remedy: several Dirichlet starts or a quadratic-programming warm start (identifiability vignette)

H.2 Active null proportions

- design: $J = 3$, $G = 10$, true $p_3 = 0$, 1000 replicates; Wald, profile-likelihood and bootstrap intervals compared on coverage near the face
- on the face the delta-method covariance is undefined, and NNLS clips p_3 to zero by construction
- the boundary likelihood-ratio test uses $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ [22, 153]; several absent types sharing signal need bootstrap-calibrated weights
- covariance input, at $\boldsymbol{p} = \mathbf{1}/J$: true full $\boldsymbol{\Sigma}_j$, cell-type diagonal, global weighted diagonal, identity; mean cosine $\{0, 0.5, 0.9\}$, $G = 50$, $J = 3$, 200 replicates; expected: the gap to the oracle grows with the mean cosine (covariance-modelling vignette)

H.3 Model robustness

H.3.1 Ablation analysis

- timing: once a first relevant gene selection and a model adapted to single-cell data (Section 3.7.4) are available, run a complete ablation study
- open design question: use the Suppinger bulk profiles with the matched single-cell profiles as ground truth, or pseudo-bulks built from the Suppinger single-cell data, whose composition is known exactly [1]
- factors
  (a) gene-selection method: our analysis of deviance (Section 3.7.2), which plays the role of an ANOVA for non-Gaussian counts; a coarse selection of regulators with *GRNBoost2* [157]; and highly variable genes as a baseline [46]
  (b) gene-selection stringency: nested panels of increasing size
  (c) covariance-structure inference: the graphical lasso; a direct, dense covariance estimate that is not sparse [59]; and partial correlations as a basic baseline (Section 3.7.3)
- expected result: a calibration curve, analogous to the bias–variance trade-off, with an optimum for a parsimonious number of genes; as the number of genes grows, performance stabilises or degrades, and DeCovarT no longer outperforms methods that use mean profiles only

H.3.2 Sensitivity to the reference covariance

- observation law: Gaussian baseline, multivariate t_3, log-normal or Poisson–log-normal counts [86], and a contaminated Gaussian ($\varepsilon = 0.05$)
- missing reference component (0, 2 or 10%), handled by ignoring it, renormalising, or adding a null type; metrics: RMSE of the remaining types and the false-inclusion rate
- dependence through Gaussian or Clayton copulas with negative-binomial margins, Kendall's $\tau \in \{0, 0.2, 0.5\}$, mirroring a HADACA3 scenario [106]
- spillover between types, and a reference covariance estimated from 20, 50 or 200 donors
- outlier genes (10 to 30%), dropout and protocol shift between reference and bulk [5]
- status: planned

Glossary

ambient likelihood and Hessian the log-likelihood $\ell(\boldsymbol{p})$ and its derivatives taken with respect to the J proportions as if they were free coordinates of $(0,1)^J$, ignoring $\sum_j p_j = 1$; the constrained derivatives follow by the chain rule through a chart

chart a local coordinate map: a smooth bijection between an open subset of $\mathbb{R}^d$ and a manifold, here between $\mathbb{R}^{J-1}$ and the open simplex, whose inverse is also smooth; optimising in a chart turns a constrained problem into an unconstrained one

closure the division of a vector of positive values by its total, as in TPM or CPM normalisation; the result lies on a simplex and its entries are no longer free

complementary slackness the KKT condition $\nu_j p_j = 0$ pairing each inequality constraint with its multiplier: either the constraint is inactive ($p_j > 0$) and its multiplier is zero, or it binds exactly ($p_j = 0$) and its multiplier may be positive; it identifies the constraints that determine the optimum

condition number $\kappa(\boldsymbol{X}) = \lambda_{\max}/\lambda_{\min}$, the ratio of the largest to the smallest eigenvalue of a symmetric positive-definite matrix; a large value signals strong multicollinearity, an ill-posed inversion and a loss of numerical precision

convolution the law of a sum of independent random vectors; the sum of independent Gaussian vectors is Gaussian, with summed means and summed covariances

diag *operator* the operator that keeps the diagonal entries of a matrix and sets the off-diagonal entries to zero; applied to a vector, it builds the diagonal matrix with that vector on its diagonal

exogeneity the regression assumption that the regressors are fixed, or independent of the error term, and that the law of the error does not depend on the regression coefficients; in deconvolution, the reference signature plays the role of the regressors and the proportions that of the coefficients

Helmert basis an explicit orthonormal basis of the hyperplane $\mathbf{1}_J^{\perp}$ of vectors summing to zero; column j is $(1, \dots, 1, -j, 0, \dots, 0)^{\top}/\sqrt{j(j+1)}$, the Gram–Schmidt orthonormalisation of the successive differences $\boldsymbol{e}_j - \boldsymbol{e}_{j+1}$

isometric log-ratio (ILR) chart the map $\boldsymbol{\rho} = \mathbf{V}^{\top} \log \boldsymbol{p}$, with inverse $\boldsymbol{p} = \operatorname{softmax}(\mathbf{V}\boldsymbol{\rho})$, between the open simplex and $\mathbb{R}^{J-1}$; $\mathbf{V}$ is a Helmert basis orthogonal to $\mathbf{1}$

isometry a map that preserves distances; the ILR chart is an isometry between the simplex with the Aitchison distance and $\mathbb{R}^{J-1}$ with the Euclidean distance, so a given fold change between two cell types has the same length whatever their abundance

Karush–Kuhn–Tucker (KKT) conditions the first-order conditions met by a local optimum of a problem with equality and inequality constraints under a constraint qualification: stationarity of the Lagrangian, feasibility, non-negative multipliers for the inequalities, and complementary slackness; linear constraints, such as those of the simplex, always qualify

overlap the MixSim misclassification probability of two Gaussian components under the Bayes rule, averaged over pairs of cell types

positive definite a symmetric matrix $\boldsymbol{A}$ with $\boldsymbol{x}^{\top}\boldsymbol{A}\boldsymbol{x} > 0$ for every non-zero $\boldsymbol{x}$; a covariance must be positive definite for the Gaussian density to exist

precision matrix the inverse $\mathbf{\Omega} = \mathbf{\Sigma}^{-1}$ of a covariance matrix; for a Gaussian, a zero entry $\omega_{gg'} = 0$ means that genes g and g' are independent given all other genes

unit simplex the set $\Delta^{J-1} = \{\boldsymbol{p} \in [0,1]^J : \sum_j p_j = 1\}$ of the compositions of J cell types

spectrum the set of eigenvalues of a square matrix

References

[1] Suppinger, S. *et al.* Multimodal characterization of murine gastruloid development. *Cell Stem Cell* **30** (2023).

[2] Newman, A. *et al.* Robust enumeration of cell subsets from tissue expression profiles. *Nature methods* **12** (2015).

[3] Newman, A. M. *et al.* Determining cell type abundance and expression from bulk tissues with digital cytometry. *Nature Biotechnology* **37** (2019).

[4] Gong, T. & Szustakowski, J. D. DeconRNASeq: A statistical framework for deconvolution of heterogeneous tissue samples based on mRNA-Seq data. *Bioinformatics (Oxford, England)* **29** (2013).

[5] Wang, X., Park, J., Susztak, K., Zhang, N. R. & Li, M. Bulk tissue cell type deconvolution with multi-subject single-cell expression reference. *Nature Communications* **10** (2019).

[6] Tsoucas, D. *et al.* Accurate estimation of cell-type composition from gene expression data. *Nature Communications* **10** (2019).

[7] Jew, B. *et al.* Accurate estimation of cell composition in bulk expression through robust integration of single-cell information. *Nature Communications* **11** (2020).

[8] Aitchison, J. The statistical analysis of compositional data. *Journal of the Royal Statistical Society: Series B (Methodological)* **44**, 139–160 (1982).

[9] Dessole, M., Dell'Orto, M. & Marcuzzi, F. The Lawson-Hanson Algorithm with Deviation Maximization: Finite Convergence and Sparse Recovery. *Numerical Linear Algebra with Applications* **30**, e2490 (2023).

[10] Erdmann-Pham, D. D., Fischer, J., Hong, J. & Song, Y. S. Likelihood-based deconvolution of bulk gene expression data using single-cell references. *Genome Research* **31**, 1794–1806 (2021).

[11] Lu, S. *et al.* Transcriptome size matters for single-cell RNA-seq normalization and bulk deconvolution. *Nature Communications* **16** (2025).

[12] Cable, D. M. *et al.* Robust decomposition of cell type mixtures in spatial transcriptomics. *Nature Biotechnology* **40**, 517–526 (2022).

[13] Gynter, A., Meistermann, D., Lähdesmäki, H. & Kilpinen, H. DeconV: Probabilistic cell type deconvolution from bulk RNA-sequencing data. bioRxiv (2023).

[14] Cai, B., Zhang, E. J., Li, H., Su, C. & Zhao, H. Statistical inference of cell-type proportions estimated from bulk expression data. *Journal of the American Statistical Association* **119**, 2521–2532 (2024).

[15] Xie, D., Gui, L. & Wang, J. Statistical inference for cell type deconvolution. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **88**, 1449–1468 (2026).

[16] Erkkilä, T. *et al.* Probabilistic analysis of gene expression measurements from heterogeneous tissues. *Bioinformatics* **26** (2010).

[17] Andrade Barbosa, B. *et al.* Bayesian log-normal deconvolution for enhanced in silico microdissection of bulk gene expression data. *Nature Communications* **12** (2021).

[18] Ghaffari, S. *et al.* BEDwARS: A robust Bayesian approach to bulk gene expression deconvolution with noisy reference signatures. *Genome Biology* **24**, 178 (2023).

[19] Chu, T., Wang, Z., Pe'er, D. & Danko, C. G. Cell type and gene expression deconvolution with BayesPrism enables bayesian integrative analysis across bulk and single-cell rna sequencing in oncology. *Nature Cancer* **3**, 505–517 (2022).

[20] Chassagnol, B., Nuel, G. & Becht, E. DeCovarT, a multidimensional probabilistic model for the deconvolution of heterogeneous transcriptomic samples. arXiv (2023).

[21] Egozcue, J. J., Pawlowsky-Glahn, V., Mateu-Figueras, G. & Barceló-Vidal, C. Isometric Logratio Transformations for Compositional Data Analysis. *Mathematical Geology* **35**, 279–300 (2003).

[22] Self, S. G. & Liang, K.-Y. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association* **82**, 605–610 (1987).

[23] Melnykov, V., Chen, W.-C. & Maitra, R. MixSim: An R Package for Simulating Data to Study Performance of Clustering Algorithms. *Journal of Statistical Software* **51** (2012).

[24] Riani, M., Cerioli, A., Perrotta, D. & Torti, F. Simulating mixtures of multivariate data with fixed cluster overlap in FSDA library. *Advances in Data Analysis and Classification* **9**, 461–481 (2015).

[25] Morris, T. P., White, I. R. & Crowther, M. J. Using simulation studies to evaluate statistical methods. *Statistics in Medicine* **38**, 2074–2102 (2019).

[26] Barabási, A.-L. & Albert, R. Emergence of Scaling in Random Networks. *Science* **286** (1999).

[27] Holland, P. W., Laskey, K. B. & Leinhardt, S. Stochastic blockmodels: First steps. *Social Networks* **5** (1983).

[28] Chiquet, J., Mariadassou, M. & Robin, S. Variational inference for sparse network reconstruction from count data. arXiv (2018).

[29] Tsagris, M. Constrained least squares simplicial-simplicial regression. *Statistics and Computing* **35**, 27 (2025).

[30] Friedman, J., Hastie, T. & Tibshirani, R. Sparse inverse covariance estimation with the graphical lasso. *Biostatistics (Oxford, England)* **9** (2008).

[31] Gilbert, P. & Varadhan, R. *numDeriv: Accurate Numerical Derivatives* (2019). URL http://optimizer.r-forge.r-project.org/. R package version 2016.8-1.1.

[32] Marquardt, D. W. An Algorithm for Least-Squares Estimation of Nonlinear Parameters. *Journal of the Society for Industrial and Applied Mathematics* **11** (1963).

[33] Philipps, V., Hejblum, B. P., Prague, M., Commenges, D. & Proust-Lima, C. Robust and Efficient Optimization Using a Marquardt-Levenberg Algorithm with R Package marqLevAlg. *The R Journal* **13** (2021).

[34] Commenges, D., Jacqmin-Gadda, H., Proust, C. & Guedj, J. A Newton-Like Algorithm for Likelihood Maximization: The Robust-Variance Scoring Algorithm. arXiv (2006).

[35] Pawlowsky-Glahn, V., Egozcue, J. J. & Tolosana-Delgado, R. in *Coordinate representation Modelling and Analysis of Compositional Data* 32–64 (John Wiley & Sons, Ltd, 2015).

[36] Boyd, S., Boyd, S. P. & Vandenberghe, L. *Convex Optimization* 1 edn (Cambridge University Press, 2004).

[37] Oehlert, G. W. A note on the delta method. *The American Statistician* **46**, 27–29 (1992).

[38] Wilks, S. S. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics* **9**, 60–62 (1938).

[39] Yee, T. W. On the Hauck-Donner Effect in Wald Tests: Detection, Tipping Points, and Parameter Space Characterization. *Journal of the American*

Statistical Association **117**, 1763–1774 (2022).

[40] Watts, D. J. & Strogatz, S. H. Collective dynamics of 'small-world' networks. *Nature* **393**, 440–442 (1998).

[41] Broido, A. D. & Clauset, A. Scale-free networks are rare. *Nature Communications* **10** (2019).

[42] Maitra, R. & Melnykov, V. Simulating Data to Study Performance of Finite Mixture Modeling and Clustering Algorithms. *Journal of Computational and Graphical Statistics* **19**, 354–376 (2010).

[43] Besson, O. & Abramovich, Y. I. On the Fisher Information Matrix for Multivariate Elliptically Contoured Distributions. *IEEE Signal Processing Letters* **20**, 1130–1133 (2013).

[44] Zimmerman, K. D., Espeland, M. A. & Langefeld, C. D. A practical solution to pseudoreplication bias in single-cell studies. *Nature Communications* **12**, 738 (2021).

[45] Squair, J. W. *et al.* Confronting false discoveries in single-cell differential expression. *Nature Communications* **12** (2021).

[46] Hafemeister, C. & Satija, R. Normalization and variance stabilization of single-cell rna-seq data using regularized negative binomial regression. *Genome Biology* **20**, 296 (2019).

[47] Agresti, A. *Foundations of Linear and Generalized Linear Models* (Wiley, 2015).

[48] Lund, S. P., Nettleton, D., McCarthy, D. J. & Smyth, G. K. Detecting Differential Expression in RNA-sequence Data Using Quasi-likelihood with Shrunken Dispersion Estimates. *Statistical Applications in Genetics and Molecular Biology* **11** (2012).

[49] Ahlmann-Eltze, C. & Huber, W. glmgampoi: Fitting gamma-poisson generalized linear models on single cell count data. *Bioinformatics* (2020). URL https://doi.org/10.1093/bioinformatics/btaa1009.

[50] Lause, J., Berens, P. & Kobak, D. Analytic Pearson residuals for normalization of single-cell RNA-seq UMI data. *Genome Biology* **22**, 258 (2021).

[51] Freedman, D. & Lane, D. A nonstochastic interpretation of reported significance levels. *Journal of Business & Economic Statistics* **1**, 292–298 (1983).

[52] Meinshausen, N. & Bühlmann, P. Stability selection. *Journal of the Royal Statistical Society. Series B (Statistical Methodology)* **72**, 417–473 (2010).

[53] Kim, S. *ppcor: Partial and Semi-Partial (Part) Correlation* (2015). URL https://CRAN.R-project.org/package=ppcor.

[54] Foygel, R. & Drton, M. Extended Bayesian Information Criteria for Gaussian Graphical Models (2010).

[55] Jankova, J. & van de Geer, S. Confidence intervals for high-dimensional inverse covariance estimation. *Electronic Journal of Statistics* **9** (2015).

[56] Liu, H., Lafferty, J. & Wasserman, L. The nonparanormal: Semiparametric estimation of high dimensional undirected graphs. *Journal of Machine Learning Research* **10**, 2295–2328 (2009).

[57] Liu, W. Gaussian Graphical Model Estimation with False Discovery Rate Control (2013).

[58] Zhang, R., Ren, Z. & Chen, W. SILGGM: An extensive R package for efficient statistical inference in large-scale gene networks. *PLOS Computational Biology* **14** (2018).

[59] Schäfer, J. & Strimmer, K. A shrinkage approach to large-scale covariance matrix estimation and implications for functional genomics. *Statistical Applications in Genetics and Molecular Biology* **4**, Article32 (2005).

[60] Chiquet, J. *Contributions to Sparse Methods for Complex Data Analysis*. Ph.D. thesis, Université Paris-Saclay / AgroParisTech (2015).

[61] Heinävaara, O., Leppä-aho, J., Corander, J. & Honkela, A. On the inconsistency of $\ell 1$-penalised sparse precision matrix estimation. *BMC Bioinformatics* **17**, 448 (2016).

[62] Sarkar, A. & Stephens, M. Separating measurement and expression models clarifies confusion in single-cell RNA sequencing analysis. *Nature Genetics* **53**, 770–777 (2021).

[63] Fang, M. & Pachter, L. Extrinsic biological stochasticity and technical noise normalization of single-cell RNA sequencing data. bioRxiv preprint (2025).

[64] Racle, J., de Jonge, K., Baumgaertner, P., Speiser, D. E. & Gfeller, D. Simultaneous enumeration of cancer and immune cell types from bulk tumor gene expression data. *eLife* **6** (2017).

[65] Wilson, E. B. Probable Inference, the Law of Succession, and Statistical Inference. *Journal of the American Statistical Association* **22**, 209–212 (1927).

[66] Zaitsev, A. *et al.* Precise reconstruction of the TME using bulk RNA-seq and a machine learning algorithm trained on artificial transcriptomes. *Cancer Cell* **40** (2022).

[67] Finotello, F. *et al.* Molecular and pharmacological modulators of the tumor immune contexture revealed by deconvolution of RNA-seq data. *Genome Medicine* **11** (2019).

[68] Monaco, G. *et al.* RNA-Seq Signatures Normalized by mRNA Abundance Allow Absolute Deconvolution of Human Immune Cell Types. *Cell Reports* **26** (2019).

[69] Avila Cobos, F., Alquicira-Hernandez, J., Powell, J., Mestdagh, P. & De Preter, K. Comprehensive benchmarking of computational deconvolution of transcriptomics data. *bioRxiv* (2020).

[70] Dong, M. *et al.* SCDC: Bulk gene expression deconvolution by multiple single-cell RNA sequencing references. *Briefings in Bioinformatics* **22** (2021).

[71] Völkl, D. *et al.* HIDE: Hierarchical cell-type deconvolution. *Bioinformatics* **41**, i207–i216 (2025).

[72] Eder, B. *et al.* Rectangle: Robust and scalable multiscale deconvolution informed by single-cell RNA sequencing data. bioRxiv (2026).

[73] Huang, P., Cai, M., Lu, X., McKennan, C. & Wang, J. Accurate estimation of rare cell-type fractions from tissue omics data via hierarchical deconvolution. *The Annals of Applied Statistics* **18** (2024).

[74] Hart, Y. *et al.* Inferring biological tasks using Pareto analysis of high-dimensional data. *Nature Methods* **12**, 233–235 (2015).

[75] Wang, Y. & Zhao, H. Non-linear archetypal analysis of single-cell RNA-seq data by deep autoencoders. *PLOS Computational Biology* **18**, e1010025 (2022).

[76] Nguyen, H. *et al.* DeOPUS: Cellular deconvolution via optimized power-transformed unmixing with shrinkage. *Briefings in Bioinformatics* **27**, bbag524 (2026).

[77] Fan, J. *et al.* MuSiC2: cell-type deconvolution for multi-condition bulk RNA-seq data. *Briefings in Bioinformatics* **23** (2022).

[78] Lau, R. *et al.* cellGeometry: Ultra-fast single-cell deconvolution of bulk RNA-Seq using a geometric solution. bioRxiv (2026).

[79] Zhang, Z. *et al.* Integrated inference of cellular compositions and gene expression programs by deconvolution. *Cell Regeneration* **15** (2026).

[80] Hampel, F. R. in *Introduction to Huber (1964) Robust Estimation of a Location Parameter Breakthroughs in Statistics: Methodology and Distribution* (Kotz, S. & Johnson, N. L.eds) 479–491 (Springer, 1992).

[81] Maronna, R. A., Martin, R. D., Yohai, V. J. & Salibián-Barrera, M. in *Generalized Linear Models Robust Statistics* 271–291 (John Wiley & Sons, Ltd, 2018).

[82] ROUSSEEUW, P. J. & VAN DRIESSEN, KATRIEN. Computing LTS Regression for Large Data Sets. *Data Mining and Knowledge Discovery* **12** (2006).

[83] Rousseeuw, P. Multivariate Estimation With High Breakdown Point. *Mathematical Statistics and Applications Vol. B* (1985).

[84] McGregor, K. *et al.* Proportionality-based association metrics in count compositional data. *NAR Genomics and Bioinformatics* **8**, lqag102 (2026).

[85] Anghel, C. V. *et al.* ISOpureR: An R implementation of a computational purification algorithm of mixed tumour profiles. *BMC Bioinformatics* **16** (2015).

[86] Chiquet, J., Mariadassou, M. & Robin, S. The poisson-lognormal model as a versatile framework for the joint analysis of species abundances. *Frontiers in Ecology and Evolution* **9** (2021).

[87] Batardière, B., Chiquet, J., Gindraud, F. & Mariadassou, M. Zero-inflation in the multivariate poisson lognormal family. *Statistics and Computing* **35** (2025).

[88] Wang, J., Roeder, K. & Devlin, B. Bayesian estimation of cell-type-specific gene expression per bulk sample with prior derived from single-cell data. Tech. Rep., openRxiv (2020).

[89] Su, C., Zhang, J. & Zhao, H. Estimating Cell-Type-Specific Gene Co-Expression Networks from Bulk Gene Expression Data with an Application to Alzheimer's Disease. *Journal of the American Statistical Association* **119**, 811–824 (2024).

[90] Tang, D., Park, S. & Zhao, H. NITUMID: Nonnegative matrix factorization-based Immune-TUmor MIcroenvironment Deconvolution. *Bioinformatics* **36** (2020).

[91] Hoyer, P. O. Non-negative Matrix Factorization with Sparseness Constraints. *The Journal of Machine Learning Research* **5**, 1457–1469 (2004). URL https://www.jmlr.org/papers/v5/hoyer04a.html.

[92] Li, S., Xu, S., Wang, X., Ertekin-Taner, N. & Chen, D. An augmented GSNMF model for complete deconvolution of bulk RNA-seq data. *Mathematical biosciences and engineering : MBE* **22**, 988–1018 (2025).

[93] Chen, L. CAMTHC: Convex Analysis of Mixtures for Tissue Heterogeneity Characterization. https://rdrr.io/bioc/CAMTHC/ (2019).

[94] Chen, L. *et al.* swCAM: Estimation of subtype-specific expressions in individual samples with unsupervised sample-wise deconvolution. *Bioinformatics* **38**, 1403–1410 (2022).

[95] Menden, K. *et al.* Deep learning–based cell composition analysis from tissue expression profiles. *Science Advances* **6** (2020).

[96] Oku, A., Geiger, H., Robine, N. & Fu, R. DECONVersation: Single Cell Foundation Model-Derived Embeddings for Robust Cell Type Deconvolution of Bulk RNA-seq (2026).

[97] Dempster, A. P., Laird, N. M. & Rubin, D. B. Maximum Likelihood from Incomplete Data Via the *EM* Algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)* **39** (1977).

[98] Kim, Y. *et al.* Longitudinal profiling of low-abundance strains in microbiomes with ChronoStrain. *Nature Microbiology* **10**, 1184–1197 (2025).

[99] Badia-i Mompel, P. *et al.* Gene regulatory network inference in the era of single-cell multi-omics. *Nature Reviews Genetics* **24**, 739–754 (2023).

[100] Maizels, R. J. & Briscoe, J. Gene regulatory networks: From correlative models to causal explanations. *Nature Reviews Genetics* **27** (2026).

[101] Nourisa, J. *et al.* genernib: A living benchmark for gene regulatory network inference. Preprint, bioRxiv (2025).

[102] Tai, A.-S., Tseng, G. C. & Hsieh, W.-P. BayICE: A bayesian hierarchical model for semireference-based deconvolution of bulk transcriptomic data. *The Annals of Applied Statistics* **15** (2021).

[103] Montierth, M. D. *et al.* Deconvolution of Sparse-count RNA Sequencing Data for Tumor Cells Using Embedded Negative Binomial Distributions. bioRxiv (2025).

[104] Heiling, H. M., Wilson, D. R., Rashid, N. U., Sun, W. & Ibrahim, J. G. Estimating cell type composition using isoform expression one gene at a time. *Biometrics* **79**, 854–865 (2023).

[105] Chen, Z. J., Rahmani, E. & Halperin, E. Unico: a unified model for cell-type resolution genomics from heterogeneous omics data. *Genome Biology* **26** (2025).

[106] Barbot, H. & Richard, M. On the Promises and Limits of Multimodal Integration for Deconvolution: The HADACA3 Benchmark. *NeurIPS* (2026).

[107] Cai, M. *et al.* Robust and accurate estimation of cellular fraction from tissue omics data via ensemble deconvolution. *Bioinformatics* **38** (2022).

[108] Law, C. W., Chen, Y., Shi, W. & Smyth, G. K. Voom: Precision weights unlock linear model analysis tools for RNA-seq read counts. *Genome Biology* **15** (2014).

[109] Liu, G., Liu, X. & Ma, L. DecOT: Bulk deconvolution with optimal transport loss using a single-cell reference. *Frontiers in Genetics* **13** (2022).

[110] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria (2026). URL https://www.R-project.org/.

[111] Philipps, V. *et al. marqLevAlg: A Parallelized General-Purpose Optimization Based on Marquardt-Levenberg Algorithm* (2023). R package version 2.0.8.

[112] Byrd, R. H., Lu, P., Nocedal, J. & Zhu, C. A Limited Memory Algorithm for Bound Constrained Optimization. *SIAM Journal on Scientific Computing* **16**, 1190–1208 (1995).

[113] Bates, D., Maechler, M. & Jagan, M. *Matrix: Sparse and Dense Matrix Classes and Methods* (2026). URL https://codeberg.org/jaganmn/Matrix. R package version 1.7-6.

[114] Ripley, B. & Venables, B. *MASS: Support Functions and Datasets for Venables and Ripley's MASS* (2026). URL http://www.stats.ox.ac.uk/pub/MASS4/. R package version 7.3-66.

[115] Soetaert, K., Van den Meersche, K. & van Oevelen, D. *limSolve: Solving Linear Inverse Models* (2026). R package version 2.0.3.

[116] Genz, A., Bretz, F., Miwa, T., Mi, X. & Hothorn, T. *mvtnorm: Multivariate Normal and t Distributions* (2026). URL https://codeberg.org/thothorn/mvtnorm. R package version 1.4-2.

[117] Csárdi, G. *et al. igraph: Network Analysis and Visualization* (2026). URL https://r.igraph.org/. R package version 2.3.3.

[118] Melnykov, V., Chen, W.-C. & Maitra, R. *MixSim: Simulating Data to Study Performance of Clustering Algorithms* (2024). R package version 1.1-8.

[119] Hofert, M. & Lemieux, C. *qrng: (Randomized) Quasi-Random Number Generators* (2026). URL https://CRAN.R-project.org/package=qrng. R package version 0.0-11.

[120] Jiang, H. *et al. huge: High-Dimensional Undirected Graph Estimation* (2026). URL https://github.com/Gatech-Flash/huge. R package version 2.0.1.

[121] Vaughan, D., Bengtsson, H. & Dancho, M. *furrr: Apply Mapping Functions in Parallel using Futures* (2026). URL https://github.com/futureverse/furrr. R package version 0.4.0.

[122] Wickham, H. *et al. ggplot2: Create Elegant Data Visualisations Using the Grammar of Graphics* (2026). URL https://ggplot2.tidyverse.org. R package version 4.0.3.

[123] Kay, M. *ggdist: Visualizations of Distributions and Uncertainty* (2025). URL https://mjskay.github.io/ggdist/. R package version 3.3.3.

[124] Gu, Z. *ComplexHeatmap: Make Complex Heatmaps* (2026). URL https://bioconductor.org/packages/ComplexHeatmap. R package version 2.28.0.

[125] Wilke, C. O. *cowplot: Streamlined Plot Theme and Plot Annotations for ggplot2* (2025). URL https://wilkelab.org/cowplot/. R package version 1.2.0.

[126] Carter, J. S. Existence of penalised likelihood estimates and posterior propriety of separable prior distributions for Gaussian precision matrices. https://arxiv.org/abs/2505.20005v2 (2025).

[127] Dahl, J., Roychowdhury, V. & Vandenberghe, L. Maximum likelihood estimation of gaussian graphical models: Numerical implementation and topology selection. Tech. Rep., University of California, Los Angeles (2005).

[128] Mazumder, R. & Hastie, T. The Graphical Lasso: New Insights and Alternatives. *Electronic Journal of Statistics* **6** (2011).

[129] Carter, J. S. & Molinari, C. Existence and optimisation of the partial correlation graphical lasso (2025).

[130] Peng, J., Wang, P., Zhou, N. & Zhu, J. Partial Correlation Estimation by Joint Sparse Regression Models. *Journal of the American Statistical Association* **104**, 735–746 (2009).

[131] Zhang, T. & Zou, H. Sparse precision matrix estimation via lasso penalized D-trace loss. *Biometrika* **101**, 103–120 (2014).

[132] Chiquet, J., Grandvalet, Y. & Ambroise, C. Inferring Multiple Graphical Structures. *Statistics and Computing* **21**, 537 (2011).

[133] Danaher, P., Wang, P. & Witten, D. M. The joint graphical lasso for inverse covariance estimation across multiple classes (2012).

[134] Chiquet, J., Grandvalet, Y. & Charbonnier, C. Sparsity with sign-coherent groups of variables via the cooperative-Lasso. *The Annals of Applied Statistics* **6** (2012).

[135] Yuan, H., Xi, R., Chen, C. & Deng, M. Differential Network Analysis via the Lasso Penalized D-Trace Loss (2017).

[136] Liu, H., Roeder, K. & Wasserman, L. Stability approach to regularization selection (StARS) for high dimensional graphical models. *Advances in Neural Information Processing Systems* **24**, 1432–1440 (2010).

[137] Laszkiewicz, M., Fischer, A. & Lederer, J. Thresholded Adaptive Validation: Tuning the Graphical Lasso for Graph Recovery. *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics* 1864–1872 (2021).

[138] Li, Y., Craig, B. A. & Bhadra, A. The Graphical Horseshoe Estimator for Inverse Covariance Matrices (2019).

[139] Newman, M. E. J. Random graphs as models of networks. arXiv (2002). URL https://arxiv.org/abs/cond-mat/0202208.

[140] Newman, M., Barabási, A.-L. & Watts, D. J. *The Structure and Dynamics of Networks* (Princeton University Press, 2011).

[141] Lima-Mendez, G. & van Helden, J. The powerful law of the power law and other myths in network biology1. *Molecular BioSystems(MBS)* **5** (2009).

[142] Abbas, A. R., Wolslegel, K., Seshasayee, D., Modrusan, Z. & Clark, H. F. Deconvolution of blood microarray data identifies cellular activation patterns in systemic lupus erythematosus. *PloS One* **4** (2009).

[143] Shen-Orr, S. S. *et al.* Cell type–specific gene expression differences in complex tissues. *Nature Methods* **7** (2010).

[144] Lähdesmäki, H., Shmulevich, l., Dunmire, V., Yli-Harja, O. & Zhang, W. In silico microdissection of microarray data from heterogeneous cell populations. *BMC Bioinformatics* **6** (2005).

[145] Tang, C. *et al.* Cell type-specific inference from bulk RNA-sequencing data by integrating single-cell reference profiles via EPIC-unmix. *Genome Biology* **26**, 395 (2025).

[146] Xie, A. & Cui, Y. gnSPADE: Incorporating Gene Network Structures Enhances Reference-Free Deconvolution in Spatial Transcriptomics. *Advanced Intelligent Systems* **8**, e202500639 (2026).

[147] Ahn, J. *et al.* DeMix: Deconvolution for mixed cancer transcriptomes using raw measured data. *Bioinformatics (Oxford, England)* **29** (2013).

[148] Wang, Z. *et al.* Transcriptome Deconvolution of Heterogeneous Tumor Samples with Immune Infiltration. *iScience* **9** (2018).

[149] Wang, C., Lin, Y., Li, S. & Guan, J. Deconvolution from bulk gene expression by leveraging sample-wise and gene-wise similarities and single-cell RNA-seq data.

BMC Genomics **25**, 875 (2024).

[150] van der Vaart, A. W. *Asymptotic Statistics* (Cambridge University Press, 2000).

[151] Firth, D. Bias reduction of maximum likelihood estimates. *Biometrika* **80**, 27–38 (1993).

[152] Batardière, B., Chiquet, J. & Mariadassou, M. Evaluating Parameter Uncertainty in the Poisson Lognormal Model with Corrected Variational Estimators. arXiv (2024).

[153] Chernoff, H. On the distribution of the likelihood ratio. *The Annals of Mathematical Statistics* **25**, 573–578 (1954).

[154] Efron, B. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics* **7**, 1–26 (1979).

[155] Sturm, G. *et al.* Comprehensive evaluation of transcriptome-based cell-type quantification methods for immuno-oncology. *Bioinformatics (Oxford, England)* **35** (2019).

[156] Allen, M. *et al.* Raincloud plots: A multi-platform tool for robust data visualization. *Wellcome Open Research* **4**, 63 (2019). URL https://wellcomeopenresearch.org/articles/4-63.

[157] Moerman, T. *et al.* GRNBoost2 and Arboreto: Efficient and scalable inference of gene regulatory networks. *Bioinformatics* **35**, 2159–2161 (2019).

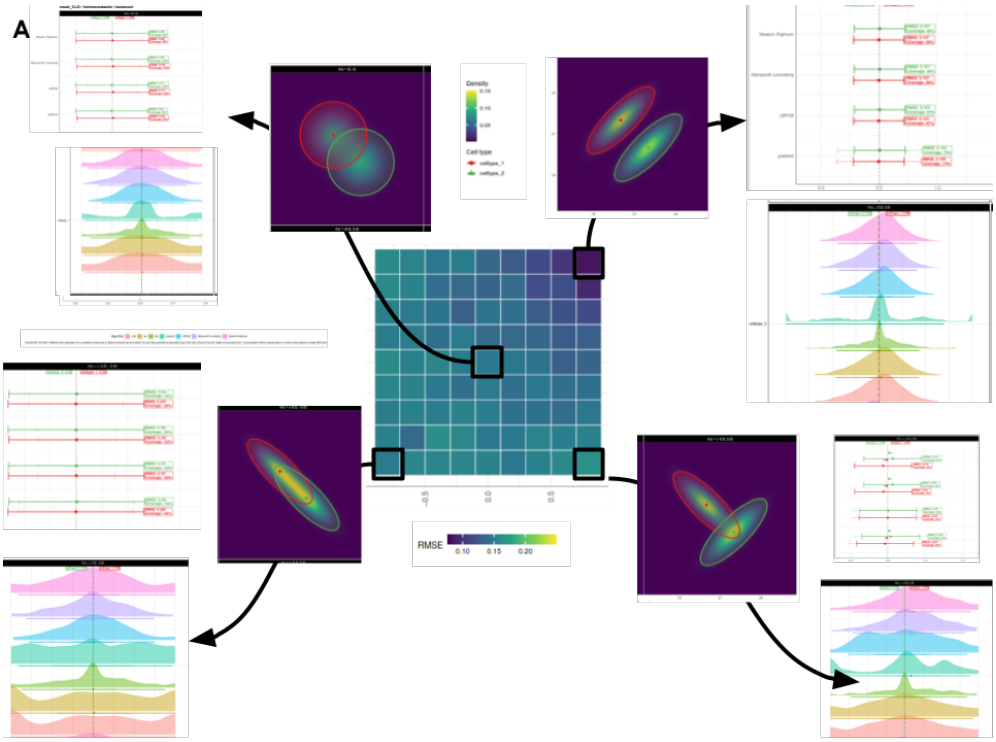


Fig. 3 Bivariate toy, panel A. Centre: root-mean-square error on the (ρ_1, ρ_2) grid (colour scale about 0.10 to 0.20). Four insets: two-gene mixture densities at correlation corners, with 95% ellipses of the purified types. Outer panels: Monte Carlo $\hat{\boldsymbol{p}}$ as dot-whisker forests and as rainclouds, for Newton–Raphson, Marquardt–Levenberg, L-BFGS, and projected gradient. Black squares and arrows mark which grid cell each density belongs to.

Incorproate the following: used the package **ComplexHeatmap** to display the mean square error (MSE) of the estimated cell ratios, comparing the NNLS output, as implemented in the DeconRNASeq algorithm ([4]), in Panel **A**, with our newly implemented DeCovarT algorithm, in Panel **B**. The lower the MSE, the least noisy and biased the estimates. In addition, we added the two-dimensional density plot for the intermediate scenario, for which each population is parameterised by a diagonal covariance matrix, and the most extreme scenarios (those with the highest correlation between genes). The ellipsoids represent for each cell population the 95% confidence region and the red spherical icon and the green triangular icon represent respectively the centroids (average expression of gene 1 and gene 2) of cell population 1 and cell population 2

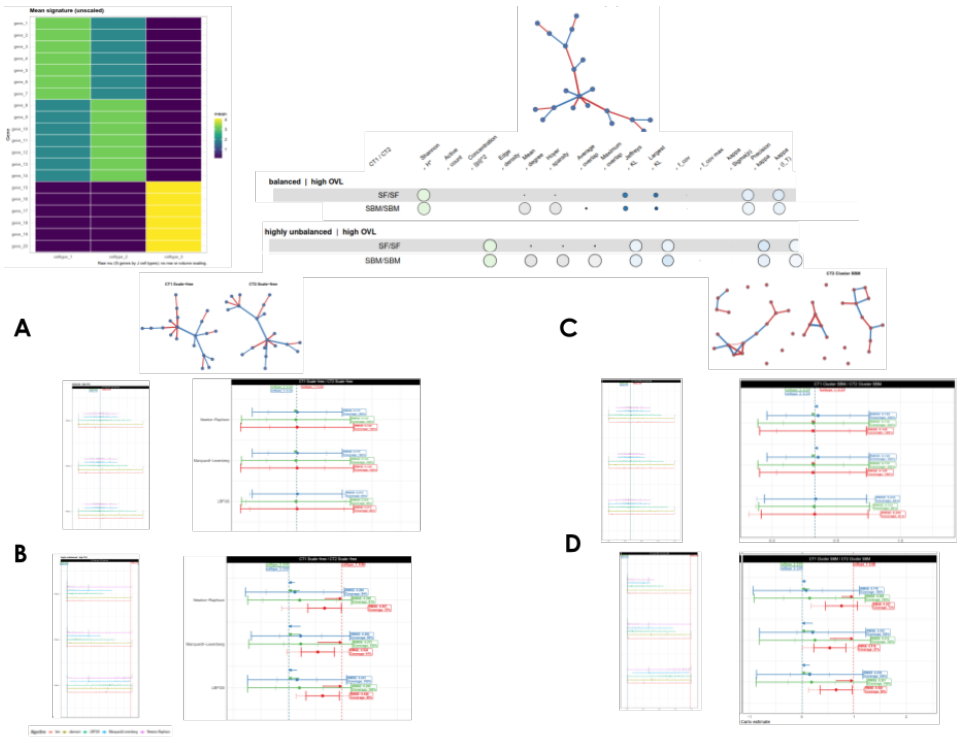


Fig. 4 Covariance-driven scenario. Top left: unscaled mean signature $\boldsymbol{\mu}$; types 1 and 2 share the high-cosine block, type 3 carries the contrasting block. Networks: scale-free and stochastic-block skeletons. Two descriptor strips, both at high overlap: balanced composition, then highly unbalanced composition (Shannon entropy, overlap, conditioning, divergences; SF/SF and SBM/SBM rows). Forests: ADEMP summaries for those two regimes. Solver colours: `lsei`, *CIBERSORT*, L-BFGS-B, Marquardt–Levenberg, Newton–Raphson. Solid whiskers use the expected-Fisher Wald standard error; dashed whiskers use the Monte Carlo standard deviation.